

UNIVERSITY OF TRIPOLI
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**Thesis Submitted In Partial Fulfillment of The Requirements For
The Degree of Master of Science In Electrical Engineering**

M.Sc. Thesis

**Study Of The Behavior Of Grid-
Connected Inverters For Photovoltaic
Systems**

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Abstract

The main objective of this thesis is to study the sudden increase of total harmonic distortion (THD) during the early morning and sunset of the 24 kW solar grid connected system to the solar systems located in the Tripoli University.

Measurements have been undertaken at the Grid Connected Photovoltaic system to the solar systems located in the Tripoli University to present and evaluate the measurements based on power quality quantities which are obtained from the PV grid-connected system site. The measured power quality parameters at the inverter output side are apparent, active and reactive powers, current, voltage and power factor. The total harmonic distortion (THD) for the current has also been measured as well, over a period of one week.

Results are presented and analyzed in order to determine how power quality quantities are affected by changes in solar irradiance .Specially During the early morning and sunset, the variation of THDi has been studied to determine the percentages of energy injected to the grid in several ranges of THDi. the obtained results show that, in general, during daylight hours 90% of the generated energy is injected to the grid with values of THDi below 5%. On the other hand, during Early morning and the sunset the obtained values point out that a small percentage of this energy is injected to the grid with values of THDi above 35%.

Using MATLAB Simulation result of the new proposed compensation technique showed very satisfactory results providing highest reduction in the current harmonics.

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DEDICATION

*This thesis is dedicated
To
My father and mother
MY Wife and sons
and
To the rest of my Family*

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LIST OF SYMBOLS AND ABBREVIATIONS

PV	Photovoltaic
SVPWM	Space vector pulse width modulation
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracking
PWM	Pulse Width Modulation
PQ	Power Quality
PCC	Point of common coupling
AI	Anti-islanding
V_{mpp}	Maximum Power Point Voltage [V]
I_{mpp}	Maximum Power Point Current [A]
P_{max}	maximum output power under given condition
q	elementary electron charge (1.602×10^{-19} C)
k	Boltzmann constant 1.381×10^{-23} (J/K)
V_d	diode voltage,
I_{pv}	PV Current [A]
I_{SC}	Short Circuit Current [A]
I_{ph}	currents generated by the solar cells (A)
I_d	Diode Current [A]
V_{OC}	Open Circuit Voltage [V]
R_s	Series resistor(Ω)
R_p	parallel resistor (Ω)
G_a	irradiance from the sunlight (W/m ²)
T	cell temperature (K)
m	modulation index
$T_{1,nom}$	nominal temperature of cell from datasheet at STC
G_{nom}	irradiance from datasheet at STC
VSI	Voltage Source Inverter
THD	Total Harmonic Distortion
DTD	dead-time distortion
AM	air mass
FFT	Fast Fourier Transform
STC	Standard Testing Conditions
IEEE	Institute of Electrical and Electronics Engineers
IEC	International Electro technical Commission

CHAPTER ONE

INTRODUCTION

1.1 General

The aim of this chapter is to present the background and motivation behind the work done in this thesis. Also, it provides the main objectives of the research as well as thesis organization.

1.2 Background

In the last two decades, increasing concerns about depletion of fossil fuel and environmental issues, mainly air pollution, led to an increasing diffusion of renewable energy sources and to continuous improvement in the technology of electrical installations to deliver renewable energy to the grid. In this context, photovoltaic (PV) systems have been one of the most developed solutions, being attractive from the political and commercial points of view. Several studies and research have also focused on the effects of harmonics in grid connected PV system and their treatment methods.

In the year 2011 the new PV production installed worldwide has been as high as about 30 GW, with a growth of 74% with respect to the previous years, reaching a total installed capacity of about 70 GW [1]. Several feed-in tariffs are now available for PV systems, thus the maximization of the productivity is very important. This goal can be achieved by solar cell technologies with high efficiency, sun-tracking systems, proper cooling techniques for PV modules in building integrated applications, master-slave control for the inverters in large grid-connected PV plants. The growth of the PV production, as well as of the production from other sources such as wind systems, has reached a level that could significantly affect the behavior of the electrical grid. For instance, in the occurrence of a disturbance in the electrical grid, the practice indicated by the early standards on distributed generation and resources, was to switch the local generation off in a short time (1s). However, with the growth of the local generation, the fast shutdown of a significant portion of local generation could

cause transient phenomena in the network, increasing the risk of grid instability. In order to overcome this problem, recent standards established that under certain voltage and frequency variations the local generation has to remain connected to the grid. In normal conditions, the PV system operation is affected by the uncertainty of the solar irradiance, temperature and local phenomena such as solar cells shading due to the location of the solar arrays and to passing clouds. In these conditions, the main aspects refer to power quality. The overall waveform quality depends on the nature of the generated power and on the structure and control of the converters that connect the PV systems to the grid. In particular, in conditions of low irradiance “specially at early morning or sunset”, the waveform of the current injected in the grid becomes heavily distorted, anyway this high distortion is associated to relatively low values of the injected current and as such it does not cause particular problems for the individual installation. The growth of the installed power in PV applications has led to solutions in which a number of inverters are connected to the point of common coupling with different topologies.

With this background and limited literature discussed of large-scale entry of PV in the Libyan power system. Techno-economics feasibility of introducing large-scale grid-connected PV power plants (PV MODEL) into the Libyan power system and the impact on the reduction of carbon dioxide emissions as it replaces to some extent conventional power plants are also presented.

1.3 Literature Review:

Many previous works deal with high penetration of PV system into power grid.

- In [1] the current harmonics generated by the PV system highly depend on the generation level. As such, conditions in which the solar irradiance is low or reduced by the effects of shadowing or passing clouds lead to an increase of the harmonic distortion. In general, at high generation level the harmonic distortion is low ($<5\%$), and the inverter control operates in such a way that

in the case analyzed the PV system is providing a positive contribution to the reduction of the distortion on the harmonic current with respect to the harmonic distortion occurring in the distribution system without the connection of the PV system.

- In [2] the power quality observations obtained from the PV grid-connected system installed in Egypt have been presented. Measurements from the PV array under test have been analyzed and evaluated to observe the overall effect of solar irradiance on the operation of the PV grid-connected system under test. Results for the two different scenarios have been considered, namely, “sunny” and “cloudy” irradiance and the effect of the solar irradiance on the power quality measurements have been investigated. It has been found that a low solar irradiance has a significant impact on the power quality of the output of the PV system.
- In [3] investigates the impact of the 10 kW residential photovoltaic (PV) system connected to the low-voltage distribution network in Croatia. The harmonic impact of such a PV plant is measured in two periods: winter and summer. Analysis of the recorded data is done regarding harmonic content of the PV current. Finally the PV plants are not the only harmonic sources in a distribution grid. Mutual influence of PV and grid harmonics is also analyzed.
- In [4] with increasing number of photovoltaic (PV) systems connected to the electricity network in Australia, power quality becomes a major concern. There is an increasing interest in Micro grids due to their potential to provide a solution to integrate renewable energy into the wider grid. This paper presents a study on the harmonic issues in a Micro grid. Different experimental scenarios have been carried out to characterize the voltage and current distortions under different operation conditions. Matlab/Simulink simulation and field experimental results have been provided. The results

showed that the emission of the harmonic current could be affected by the operational periods of PV inverters.

1.4 Motivation

Over the past few decades, the demand for renewable energy has increased significantly due to the disadvantages of fossil fuels and greenhouse effect. Among various types of renewable energy sources (RES), solar energy and wind energy have become the most promising and attractive because of advancement in power electronic technique. Photovoltaic (PV) sources are used nowadays in many applications as they own the advantage of being maintenance and pollution free. In the past few years, solar energy sources demand has grown consistently due to the following factors:

- 1- Increasing efficiency of solar cells;
- 2- manufacturing technology improvement;
- 3- economies of scale.

Furthermore, the output of PV arrays is influenced by solar irradiation and weather conditions. More importantly, high initial cost and limited life span of PV panels make it more critical to extract as much power from them as possible.

Increasing use of static power converters like rectifiers and switched mode power supplies causes injection of harmonic currents into the distribution system. Current harmonics produce voltage distortions, current distortions, and unsatisfactory operation of power systems. Therefore, harmonic mitigation plays an essential role in grid connected PV system. IEEE std. 1547 was first drafted in 1999, and later approved in 2003 and revised in 2014 to 1547-a to provide guidance dealing with Interconnecting Distributed Resources with Electric Power Systems to provide direction on dealing with harmonics produced by static power converters and nonlinear loads. This standard helps to prevent harmonics from negatively affecting the utility grid. PV inverters can be

controlled to provide harmonic currents needed by nonlinear loads so that the current on the grid side can be approximately sinusoidal using compensation technique, at least complies with the IEEE Std. 1547. This concept is simulated successfully in the simulation environment. Then, the test bed is constructed and the components required consist of: a DC power source, a DC/AC inverter, and a grid voltage supply.

1.5 Objectives:

As a result of the initial measurements carried out in the 24 kW solar grid connected system, the measurements were evaluated based on the quantities of the quality of the energy obtained from the location of the system connected to the PV network. The power quality parameters measured in the output side of the inverter such as current, voltage, active and reactive power, power factor ,and total harmonic distortion THD.

The main task of the work is to study the causes behind the sudden sharp increase of THD values during early morning and sunset and to try to propose possible solution to reduce the sudden increase in THD values. The measurements are taken at the point where the PV plant is connected to the distribution grid:

- Spring week from 6st to 13th of March.
- Summer week from 13th to 19st of June.

From The time evolution of the current THDi and plant generation with PV output power variations as can be seen,

- 1- The THDi characterizes a comparatively high value of up to 45% during the morning and evening hours when generation was low (below 10%).
- 2- Also when the generation level is higher (up to 80%) , the THDi was reduced below the value of 5%.

The compensation method suggested it seems quite most effective technique to minimize the harmonic, providing the highest reduction in harmonics.

1.6 Methodology:

This section presents the results of measurements made for the 24 kW solar grid connected system to determine the behavior and parameters of the PV inverters systems connected to the public network in order to determine the topology for modeling photovoltaic systems using PVSYST software.

PVSYST uses manufacture data to estimate a modules potential power generation. The model itself contains a wide range of input parameters including temperature coefficients, solar cell type, I-V and P-V characteristics and many other specifications making it one of the most accurate that yield assessment tools available to PV developers. The main standard conditions (STC) can be defined by a module temperature of 25°C at an irradiance level of 1000 W.m⁻² under a reference spectral distribution of air mass (AM 1.5).The models are performed in a simple network to study harmonics by power quality analyzer (Fluke 435).Finally, the network simulation is checked to study energy quality with the help of Matlab computing software to achieve the main objective.

1.7 Outline of The Thesis:

The work presented in this thesis deals with measurement and simulation of PV inverter systems. Each chapter deals with a specific area of research:

Chapter 1.Introduction:The aim of this chapter is to present the background and motivation behind the work done in this thesis. also provides the main objectives of the research as well as the thesis organization.

Chapter 2.Photovoltaic System Description: This chapter overviews the PV cell characteristics and possible modeling methods of PV modules. The chapter starts with introducing the concepts solar energy transformation to electrical energy and explaining the characteristics of PV cells.

Chapter 3. PV-Inverter Topologies: This chapter focuses on the different inverter topologies and the pulse width modulation technique used to control the gate of the transistors. There are multiple standards dealing with the whole topic

of PV Energy systems. Finally the models are validate by comparing comply with the standards.

Chapter 4. *Power Quality Field Measurements of PV Inverters:* In this chapter various measurements of grid connected inverters of small scale PV stations have been carried out during real operating conditions. Then, a comparative study of the main electrical characteristics is performed, utilizing the measurement results.

Chapter 5. *Simulation Results:* this chapter provides an our view of the system simulation of PV module with Matlab/Simulink. The chapter concludes with the model validation compared with the measurement results.

Chapter 6. *Conclusion and Future Work:* the final conclusion is presented, based on the theoretical and experimental results performed, and additional, guidelines for future work that can be done.

CHAPTER TWO

PHOTOVOLTAIC SYSTEM DESCRIPTION

2.1 General

This chapter provides an overview of different components of a photovoltaic system and attempts to highlight recent research activities in the field of photovoltaic. In order to better understand how a PV module works we have to start by the solar cell. A solar cell can be described as an electronic device able to capture the energy of the photons and directly convert it to electricity. The process is briefly described considering a crystal silicon cell, the cells used on this thesis.

2.2 How Solar PV Cells Works?

Fig.2: shows a simple circuit of a silicon solar cell with an external load:

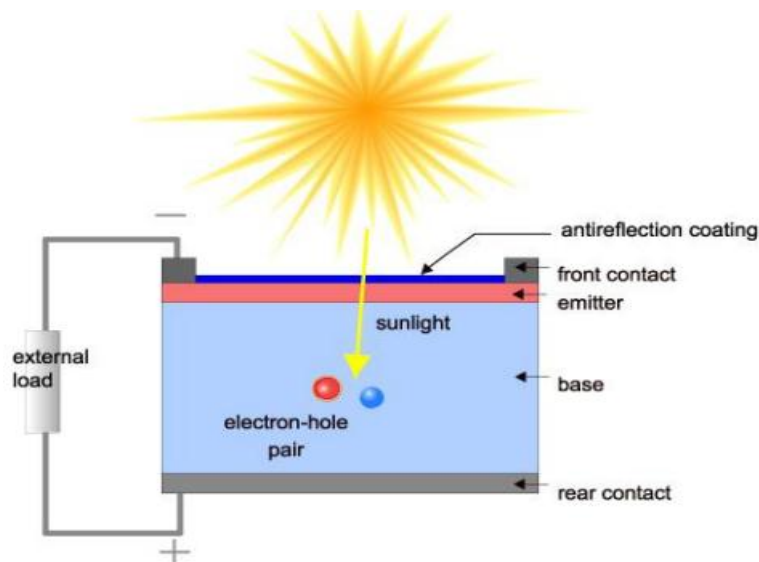


Fig.2.1: Schematic of a solar cell connected to a load

A photovoltaic cell is composed of a thin wafer consisting of an ultra-thin layer of phosphorus doped (N-type) silicon on top of a thicker layer of boron-doped (P-type) silicon. At the p-n junction near the top surface of the cell an electrical field is created. When sunlight strikes the surface of a PV cell, this electrical field provides momentum and direction to light-stimulated electrons, resulting in a flow of current when the solar cell is connected to an electrical load. A typical silicon PV cell produces about 0.5 - 0.6 volt DC under open-circuit, no-load conditions. The current (and power) output of a PV cell depends on its

efficiency and size (surface area), it is proportional to the intensity of sunlight striking the surface of the cell. The internally generated photocurrent in a solar cell is proportional to the irradiance.[5]

2.3 Categories For PV Inverters

There are a few ways in which Photovoltaic Inverters (Sometimes also called Solar Inverters) can be categorized. To begin with, different types of PV Inverters exist, depending on whether they are connected or not to the electricity grid, and in which way they interact with it [8]:

- **Stand-alone inverters**, used in independent isolated loads (such as the power supply of a household via solar modules on its roof)
- **Grid-tied inverters**, in which the generated energy is fed into the grid. The properties of the signal created by the inverter have to match those of the utility network.
- **Battery backup inverters**, this special sort of inverters are designed to help control the energy within a battery and feed with the excess charge the electricity grid. They only work in those periods of energy surplus in the battery.

In solar photovoltaic energy systems, irradiation energy coming from the sun is converted into electric energy via photovoltaic cells. These cells are often connected in series so as to form a so-called solar module. Solar modules are then sold independently and can be combined in series to form a string. Strings can at the same time be also connected in parallel to make an array.

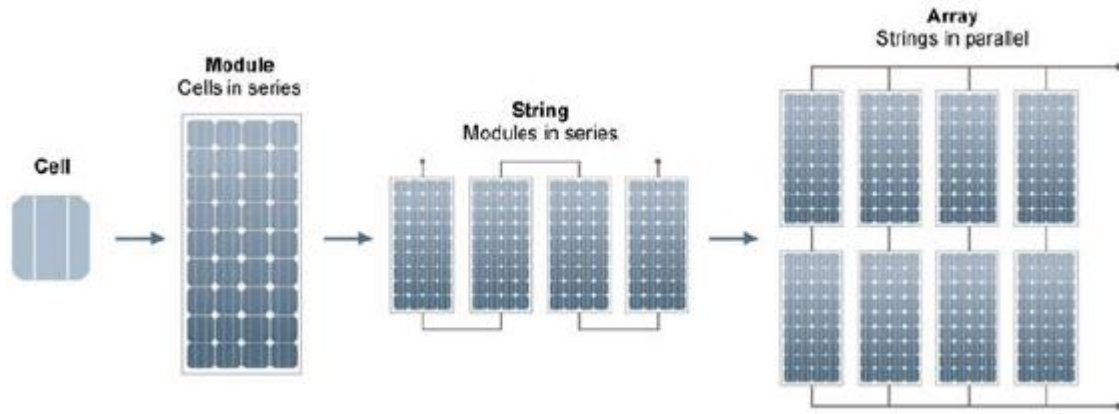


Fig.2.2: composition of string and array out of single PV modules

2.3.1 Solar Irradiance and Temperature Effects

Solar irradiance and temperature have a direct effect on the characteristics of solar PV and hence its generation and efficiency. Solar irradiation is the measure incident of total amount of solar radiation transmitted to the surface of the Earth's atmosphere in a given unit of time. It is time integral of the irradiance extended over a given period of time, therefore radiation units are units of energy ($\text{J/m}^2/\text{day}$, $\text{kWh/m}^2/\text{day}$ or $\text{kWh/m}^2/\text{month}$ [5].

Normally, the electric characteristics of a PV cell are displayed as a relation between the cell voltage and current, and a relation between the cell voltage and power. Accordingly, several electric quantities that are important to the operation of the PV system are identified. These electric quantities include: the cell voltage under open circuit conditions, V_{oc} , the cell current under short circuit conditions, I_{sc} , and the cell voltage. The effect of the irradiance on the voltage-current (V-I) and voltage-power (V-P) characteristics is depicted in Figure (2.3),(2.4), the voltage and current are normalized using the V_{oc} and the I_{sc} respectively, in order to illustrate better the effects of the irradiance on the V-I and V-P curves.

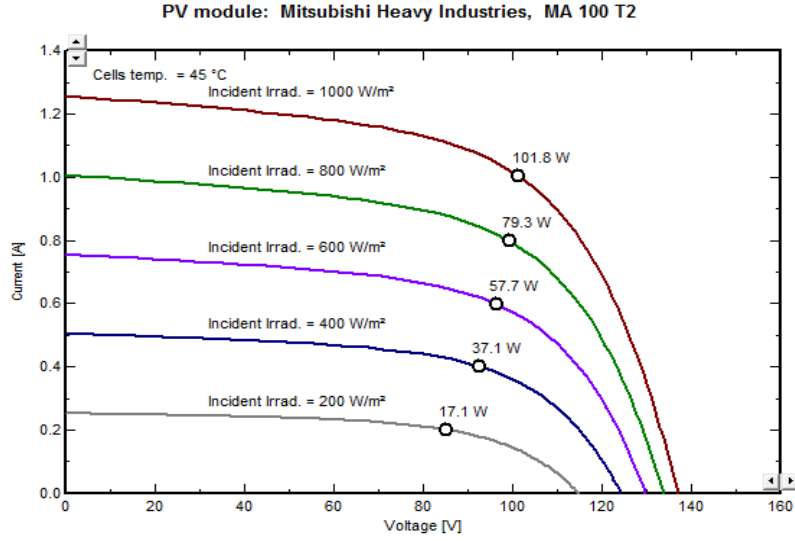


Fig.2.3: V-I curves at constant temperature (45⁰C) for different insolation values

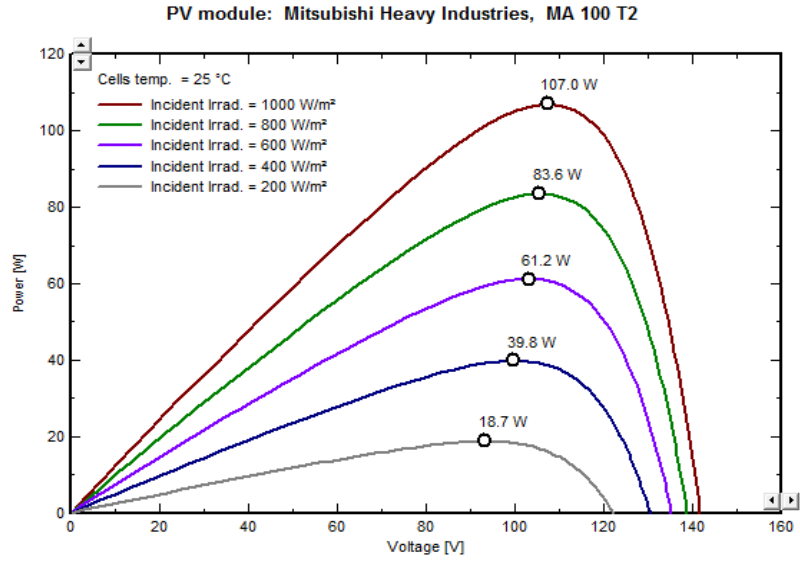


Fig.2.4: V-P curves at constant temperature (25⁰C) for different insolation values

The open circuit voltage is linearly dependent on the temperature, as shown in the equation below:

$$V_{\alpha}(T) = V_{\alpha}^{STC} + \frac{K_{v,\%}}{100}(T - 273) \quad (2.1)$$

The effect of the temperature on V_{oc} is negative, because $K_{v,\%}$ is negative, i.e., when the temperature rises, the voltage decreases. The current increases with the temperature but very little and it does not compensate the decrease in the voltage caused by a given temperature rise.

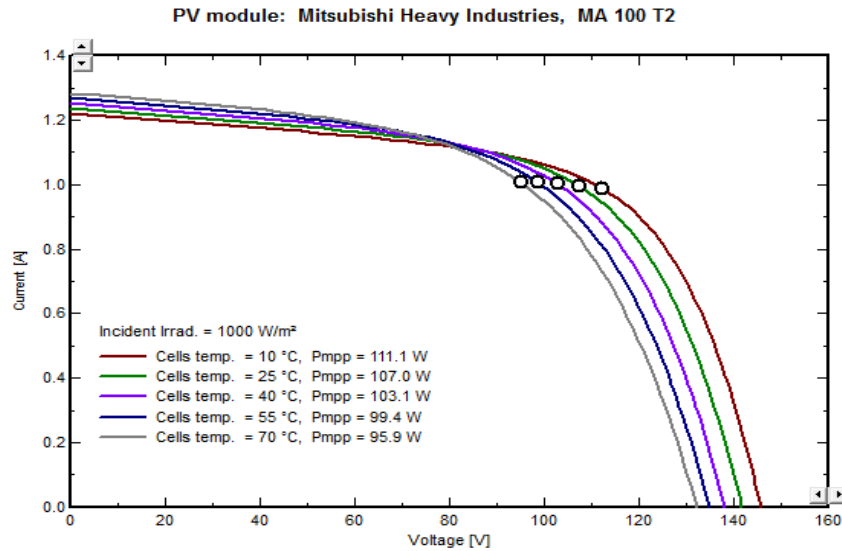


Fig.2.5: V-I curves at constant irradiation (1 kW/m²) for different temperatures

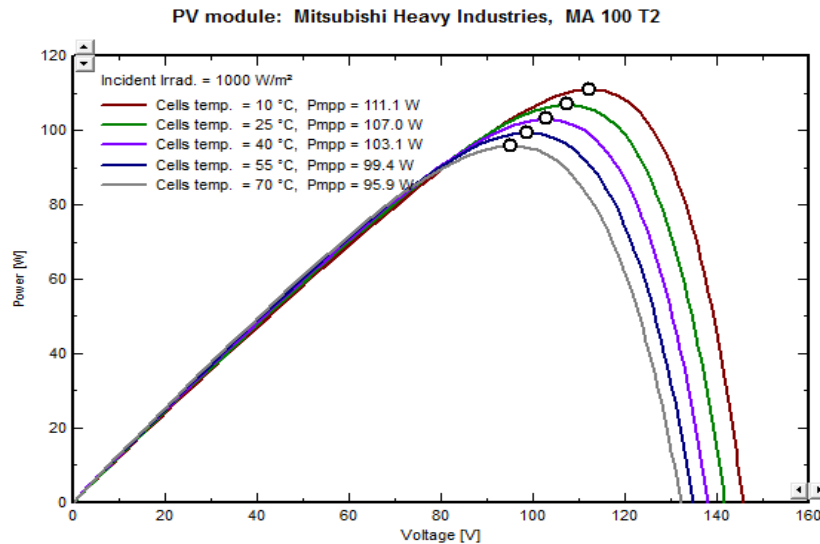


Fig.2.6: V-P curves at constant irradiation (1 KW/m²) for different temperatures

2.3.2 Peak Sun Hour

The number of peak sun hours for the day is the number of hours for which energy at the rate of 1 kW/m² would give an equivalent amount of energy to the total energy for that day. It is the equivalent number of hours per day when solar irradiance averages is 1000 w/m².

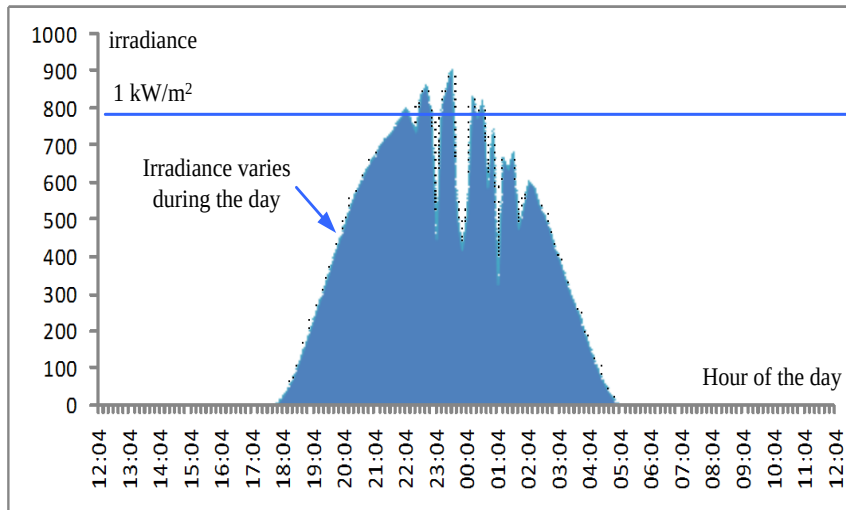


Fig.2.7: Typical variation of Solar Irradiance

This figure is not easily calculable, as it relies on Meteorological office data. Peak sun hours relate to the intensity of the sunlight or solar energy. In reality, this intensity is not actually received by the solar panel, as not all this irradiance reaches the surface of the earth. The sun hours are less important than the insolation factor, which is able to take other factors into account and give a more accurate figure of the amount of sunlight received at the Earth's surface. The weather conditions cannot be guaranteed and the sun hours quoted are only representative based on a ten year average.

2.4 Characteristic Current-Voltage (I-V)

The standard representation of the output of a PV device is called the current-voltage curve. I-V measurement determine the output performance of solar PV, including: open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), voltage at maximum power (V_{max}), and current at maximum power (I_{max}). The letter “I” is usually used to represent electrical current and the “V”

+operating voltage range. At high voltage, the relative current level rapidly falls off past a "knee" region on the curve.

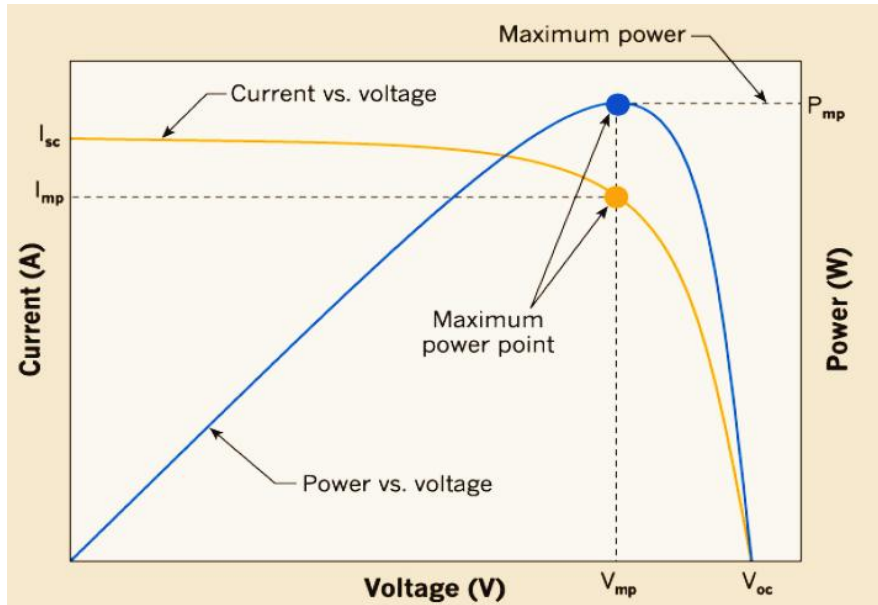


Fig.2.8: Characteristic I (V) of Solar module and power curve at STC

2.5 General Topology of Photovoltaic System

As discussed in the previous, there are several methods to connect the solar panel. The grid can be directly linked to it or the load itself. The topology of the PV system determines the type of converter interface that should be used. It depends on the configuration, the cost and the efficacy objective. In the single stage configuration, as shown in figure (2.9), the interface of the PV array to the grid is realized with the DC/AC inverter. The DC/AC inverter functions are to boost the PV voltage and control the current injected to the grid.

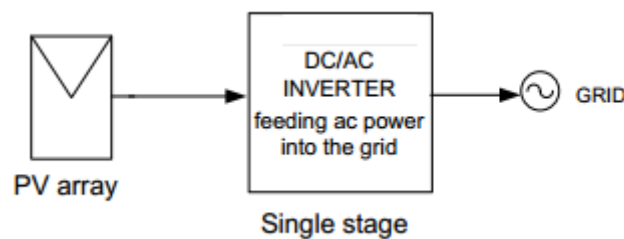


Fig.2.9: Structure of Single stage DC/AC Photovoltaic system

2.6 Modeling of PV module:

Basic approaches of PV cell modeling with Matlab/Simulink environment. and mathematical model of the PV cell are presented in this section.

2.6.1 Equivalent Circuit

A PV module consists of a number of solar cells connected in series and parallel to obtain the desired voltage and current output levels. Each solar cell is basically a p-n diode. As sunlight strikes a solar cell, the incident energy is converted directly into electrical energy without any mechanical effort.

For simplicity, the single-diode model show below is used in this thesis. This model offers a good compromise between simplicity and accuracy with the basic structure consisting of a current source with a parallel diode.

The equivalent electrical circuit is shown in Figure (2.10) Series resistors R_s and parallel (shunt) resistors R_p to limit the performance of the cell are added of the model to take into account the dissipative phenomena of the cell (internal losses) [6].

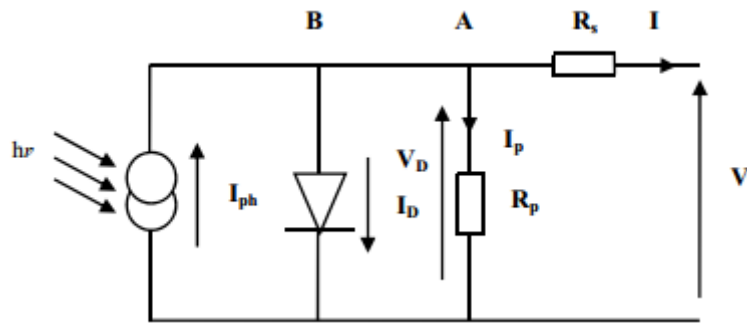


Fig.2.10 Equivalent circuit of real solar cell with R_s and R_p

2.6.2 Equation of PV Module:

PV cells are grouped in larger units called PV modules, which are further interconnected in a series-parallel configuration to form PV arrays. The following are the basic equations from the theory of semiconductors and photovoltaic that mathematically describe the I-V characteristic of the photovoltaic cell and module.

So, according to Kirchhoff law to the nodes A and B, we have:

$$I = I_{ph} - I_D - I_p \quad (2.2)$$

I_{ph} : is the light-generated current at the nominal condition (25°C and 1000W/m²), it is linearly dependent on the solar radiation and is also influenced by temperature according to the following equation:

$$I_{Ph} = [I_{Scr} + K_i(T_K - T_{ref})] \cdot \left(\frac{G}{1000}\right) \quad (2.3)$$

Where:

K_i : short-circuit current/ temperature coefficient of cell.

T_k and T_{ref} : are the working temperature of cell and reference temperature respectively in K °.

G : is the solar radiation of the cell surface (W/m²).

I_p : Current through R_p .

I_D : Diode Current, it is given by:

$$I_D = I_0 \left[\exp \left(\frac{qV_D}{AKT} \right) - 1 \right] = I_0 \left[\exp \left(\frac{q(V+R_S I)}{AKT} \right) - 1 \right] \quad (2.4)$$

Where:

q : electron charge constant ($1.6 \cdot 10^{-19}$ C),

K : Boltzmann's constant ($1.38 \cdot 10^{-23}$ J/K),

T : Cell temperature (°K).

I_0 saturation current of the diode and it is given by:

$$I_0 = I_{rs} \left[\frac{T}{T_r} \right]^3 \exp \left[\left(\frac{qE_{go}}{AK} \right) \left(\frac{1}{T_r} - \frac{1}{T} \right) \right] \quad (2.5)$$

I_{rs} : is the reverse saturation current and it is given by:

$$I_{rs} = \frac{I_{Scr}}{\exp[qV_{OC}/N_S KAT] - 1} \quad (2.6)$$

Where

A : ideality factor of the cell depends on recombination mechanisms in the space charge zone.

E_{go} : band gap energy of the semiconductor ($E_{go} \approx 1.1$ ev for the poly crystalline Si at 25°C).

Replacing in equation (2.2) the equation (2.4), the characteristic equation becomes:

$$I = N_p I_{Ph} - N_p I_0 I_0 \left[\exp \left(\frac{q(V + R_s I)}{N_s A K T} \right) - 1 \right] - \frac{V + R_s I}{R_p} \quad (2.7)$$

Where N_p and N_s are, respectively, the number of parallel and series connections of cells in the given photovoltaic module ($N_p = 1$ and $N_s = 60$).

In the ideal case, R_s tends towards 0 and R_p to infinity. In the real case, these resistors provide an assessment of the imperfections of the diode; considering that the resistance R_s has a low value, the slopes of the I-V characteristics are calculated at $I=0$ open circuit and short circuit $V=0$ and respectively give the inverse of series and shunt resistance values.

2.6.3 Physical Model of PV in Simulink

Matlab/Simulink have many ways of modeling devices or components and modeling complex systems. This section outlines three different approaches of varying degree of complexity for modeling a photovoltaic system in Matlab/Simulink, First a simple mathematical model is provided. followed by the use of advanced component libraries to design a cell. Then, a more sophisticated approach based on experimental data is provided. For the interested reader, the following reference [7] is recommended, which has been used as foundation for photovoltaic cell model implemented in this thesis.

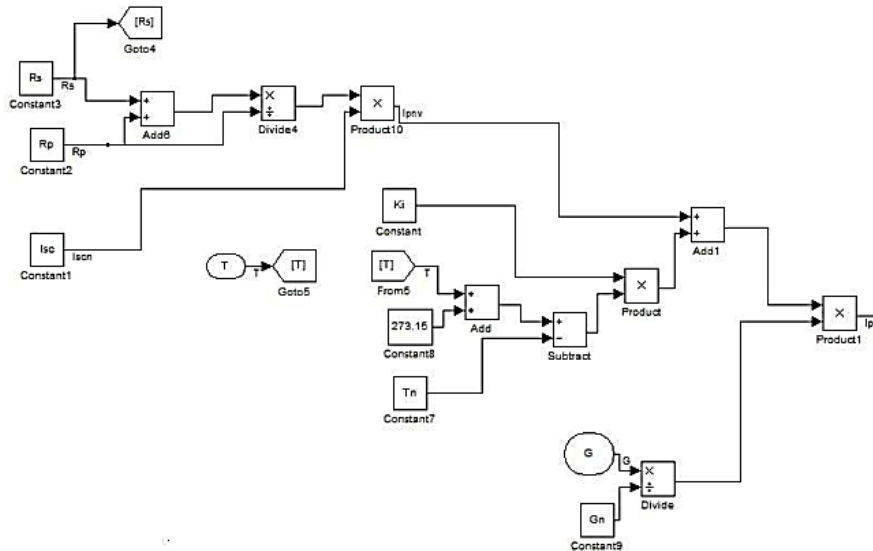


Fig.2.11: Simulink model of Ipv.

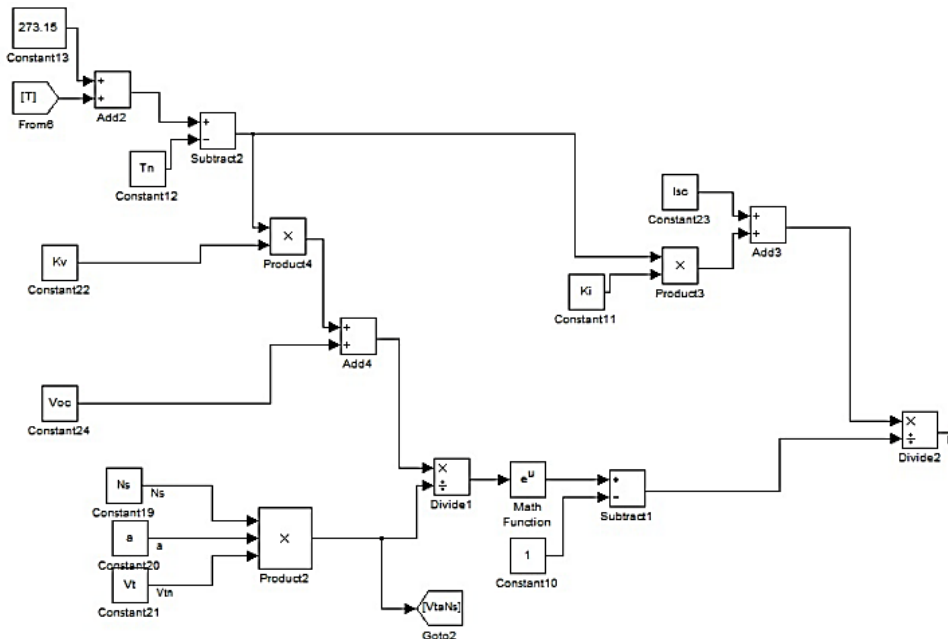


Fig.2.12: Simulink model of I_o .

The model aims to be generic and user oriented so that anyone can input the parameters of the PV module to simulation. To accomplish this subsystem was masked and a user interface inducement for the parameters found in the data sheet appears when the PV cell model is double clicked. The Figure (2.13) shows the user interface.

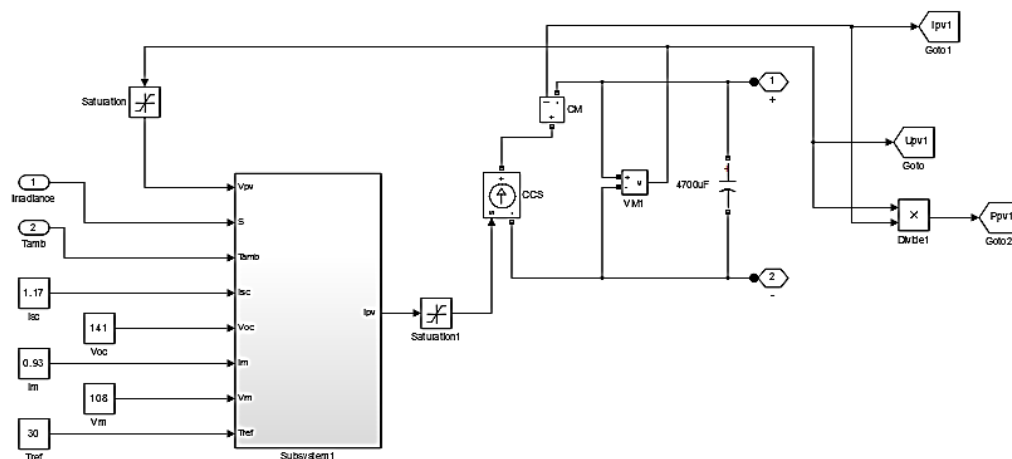


Fig.2.13: Physical model of a solar cell in Simulink

2.7 Impacts of PV Systems on the Grid:

Grid-connected PV systems might impose several negative impacts on the network, especially if their penetration level is high. Such negative impacts include power and voltage fluctuation problems, harmonic distortion, malfunctioning of protective devices and overloading and under loading of

feeders. Studying the possible impacts of PV systems on the electric network is currently becoming an important issue and is receiving a lot of attention from both researchers and electric utilities. The main reason for the importance of this issue is that accurate evaluation of these impacts, as well as providing feasible solutions for the operational problems that might arise due to installing PV systems, is considered a major contribution towards facilitating the widespread use of these systems. Due to their importance and relevance to the research presented in this thesis.

CHAPTER THREE

PV INVERTER TOPOLOGIES

3.1 General

This chapter focuses on the different inverter topologies and the pulse width modulation technique used to control the gate of the inverters which is filtered to finally produce a low-distortion sinusoidal signals (voltage and current).also this chapter discusses The multiple standards dealing with the whole topic of PV Energy systems. This standards are often written at an International (IEC) or European (EN) level and are re-issued at a national level by each country. Finally the models are validated and compared with the standards.

3.2 Inverter Topologies

The photovoltaic power systems have made a successful transition from a standalone feature to large grid-connected systems. Grid-connected PV systems can vary in size, but these systems have common components. In addition to the PV components mentioned in chapter 2,the grid connected elements necessitated to be appeared here are: the current controller and output inverter, as well as a metering equipment to monitor the power that is fed into the grid.

Grid connected PV systems have become one of the important applications of solar energy. The simple configuration of such a system and its low cost has attracted broad interest. The wide acceptance and utilization of PV power generation, in any form, are still dependent upon lowering the capital cost of system installation. Moreover, for grid connected PV plants, it makes good economic sense to maximize the amount of power generated by PV arrays and thus transferred to the grid at all times.

This will lead to the scaling down of PV arrays and converters, thus, resulting in a lower cost system. The utility grid acts as storage means for the PV arrays, as shown in figure(3.1) which will be proposed as a case study in this thesis.

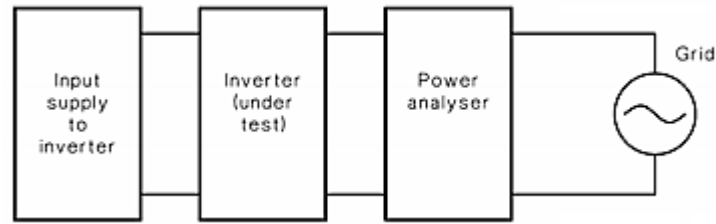


Fig. 3.1: Power circuit of a grid-connected PV system

Interconnection of power generation from PV energy sources to the grid needs DC-AC inverter. Connection of the inverters with the distribution network thought power system analyzer.

3.3 Grid Connected Solar PV Systems

Grid-connected, also known as utility-interactive PV systems are designed to operate with and inter connected to electrical utility grid. In grid-connected systems, the inverter is the primary component. The inverter converts the DC power produced by PV array into AC power consistent with the voltage and power quality requirements of the utility grid. The inverter stops the PV array from supplying power when the utility is not energized. This safety feature is required in all grid-connected PV systems, and ensures that the PV system will not continue to

operate and feed back into the utility grid when the grid is down for service or maintenance. [5].

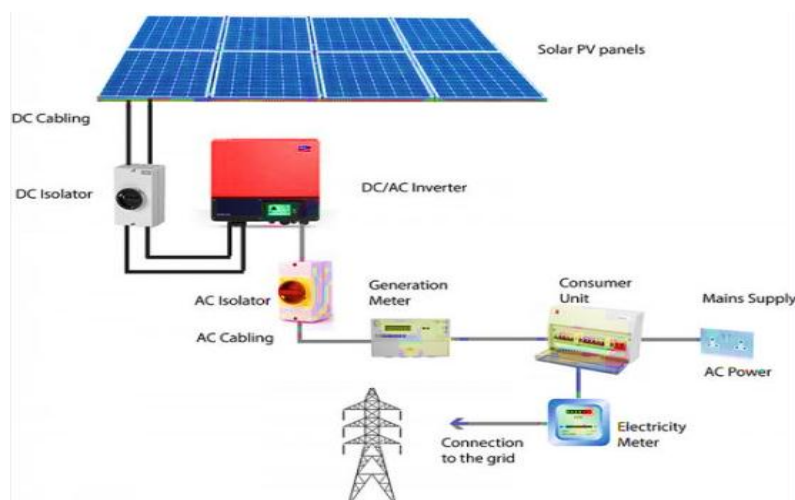


Fig.3.2: Grid-Connected system

3.3.1 Junction Box

Multiple modules are connected through a junction box, to provide surge protection.

3.3.2 DC Switch

Solar generators must be separated from the inverter in order to undertake repairs or general maintenance work on the systems. This is achieved by a DC switch installed closer to the inverter. DC switch copes with the short-circuit current as well as the open-circuit voltage as the modules continue to generate electricity. Some inverter manufacturers offer integrated DC switches. [5]

3.3.3 Inverters

In commercial capacity category, 3 phase inverters are the most suitable choice for the PV systems. The DC power coming from the photovoltaic generator is converted into a utility frequency alternating current AC by the inverter before feeding it into the electric grid. Solar inverters have some special features maximum power point tracking and anti-islanding protection.

3.4 Problems With Grid-Connected PV Systems

Obviously the benefits of the PV technology is promising, however, there are complications when integrating them with the power system [9]. The examination describes the appearance of different electrical problems that PV systems may cause on the grid. The following phenomena were measured and analyzed:

- Harmonic Distortion
- Islanding
- Distribution line short circuit
- AC-DC mixing fault
- Output fluctuation
- PV array temperature

3.4.1 Harmonic Distortion

Grid-connected PV systems need an inverter to transform DC to AC. This component causes sinusoidal disturbance in the grid since it cannot produce an ideal alternating current. This phenomenon is called harmonic distortion. When multiple inverters are connected to the grid, it has an effect on the level of harmonic distortion. Power Electronics, non-linear loads, and inverters contribute to the harmonic content within a power system. The test results shows that connecting the power electronics to the grid causes the harmonic distortion to increase in most cases [9].

3.4.2 Islanding

When the power supply from the utility station is interrupted, it occurs that the generators (e.g. PV systems) connected to the distribution line continue to provide power to the loads attached. This phenomenon is called islanding. Considering that the load supply is no longer controlled by the utility station there is a great risk of hazard for human safety but also for the equipment if islanding is not prevented. To avoid this problem there are PV-systems with anti-islanding protection. If the inverters have islanding detection, the phenomenon does not continue that long and it reduces the risk of faults and damage [9].

3.4.3 PV Array Temperature

The high temperature provides negative impacts not only on the PV modules, but also on the performance of the inverter. Thus, the effect of temperature on the inverter efficiency should be taken into account when predicting energy yield or analyzing losses of the PV systems, especially in high temperature regions.

3.5 Characteristic of Inverter

There are two common topologies of single phase inverters – full H-bridge and half-bridge. The H-bridge is superior in all specs, but it does require more

components and more complex control. Why then the use of half-bridge at all? will use the least expensive device that fulfills the requirements. In order to make a better decision, it's important to understand the differences in operation of these two types of inverters[10].

Inverters are used to convert DC voltage into AC voltage. Typical applications for inverters in power systems include interconnection of DC voltage sources to the electric grid. The focus of this thesis is the use of inverters to convert DC power generated by PV arrays to AC for interconnection to the electric grid.

3.5.1 DC/AC Inverter Analysis:

In this paragraph, inverter architectures and waveforms will be introduced as an overview. Detailed description of single-phase inverter will be analyzed for photovoltaic system. Detailed modulation strategies of the space vector modulation will be described for the single-phase inverter.

In photovoltaic system, the DC/AC inverter is used to convert the power of the source by switching the DC input voltage (or current) in a pre-determined sequence to generate AC voltage (or current) output. figure (3.3) shows the equivalent circuit of single-phase inverter. This has two switching position, turn on and off to obtain AC signal output.

figure (3.3) A representation of a simple half-bridge inverter circuit. The circuit then becomes equivalent to a resistor connected across a source of emf.

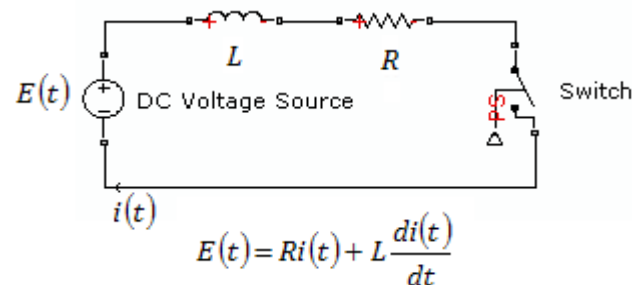


Fig.3.3. switched RL circuit and the formula describing its operation

R is the resistance of the load, L is its inductance, E(t) is the DC source voltage, i(t) is the current of the circuit. The RL circuit shown above has a resistor and

an inductor connected in series. A constant voltage $E(t)$ is applied when the switch is closed.

$$[E(t)] = R_i(t) + L \frac{di(t)}{dt} \quad (3.1)$$

By transformation the lablas transformer:

$$\text{Laplace}[E(t)] = \text{laplace} \left[R_i(t) + L \frac{di(t)}{dt} \right] \quad (3.2)$$

$$\frac{E_x}{s} = R \cdot I(s) + L(s \cdot I(s) - i(t_0)) \quad (3.3)$$

$$\frac{E_x}{s} + L \cdot i(t_0) = I(s)[R + sL] \quad (3.4)$$

$$I(s) = \frac{E_x/s}{R+sL} + \frac{L \cdot i(t_0)}{R+sL} \quad (3.5)$$

$$I(s) = \frac{E_x}{s} \cdot \frac{1}{R} \cdot \frac{1}{1+\frac{L}{R}s} + i(t_0) \frac{L}{R} \cdot \frac{1}{1+\frac{L}{R}s} \quad (3.6)$$

$$\text{Laplace}^{-1}[I(s)] = i(t) = \frac{E_x}{R} \left(1 - e^{-\frac{R}{L}t} \right) + i(t_0) \cdot e^{-\frac{R}{L}t} \quad (3.7)$$

Using the Laplace transform, it's easy to derive the formula for the current.

With this formula, there is two extreme states of the circuit:

1. $i(t_0) = 0$ and $E_X \neq 0$ (3.8)

2. $i(t_0) = E_X/R$ and $E_X = 0$ (3.9)

These two states shows (3.8,3.9) have the names of: turn-on and the turn-off of the circuit during its steady state. These formulae tell us that the current changes exponentially on turn-on and turn-off.

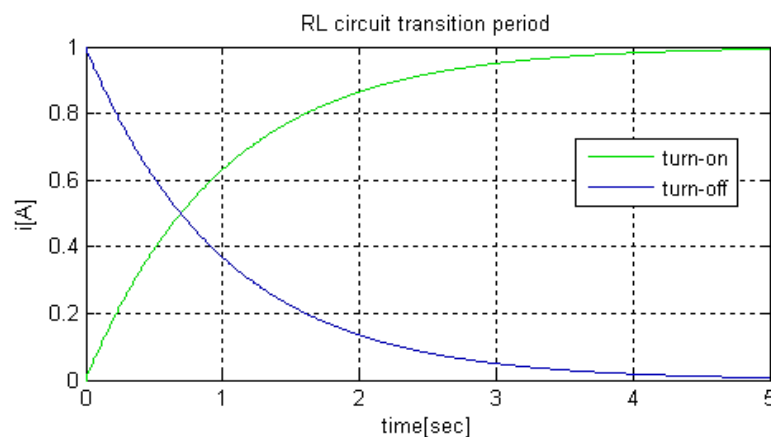


Fig.3.4. Turn-on and Turn-off of an RL circuit

It is seen that the current in an RL load is inactive, and cannot change instantaneously. Here is where the fly back diodes of the switching devices play a critical role. Let's take a close look at a inverter. For a start, let's consider the simpler example of a half-bridge circuit.

3.5.2 Half-Bridge Single Phase Inverter:

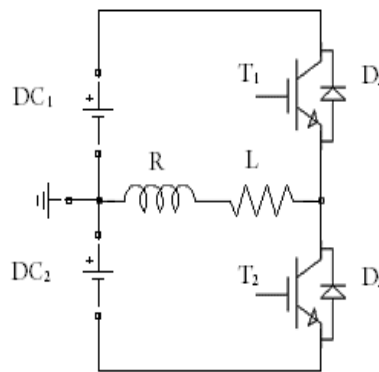


Fig.3.5. Half-Bridge circuit

Figure (3.5) shows half-bridge circuit, where DC_1 and DC_2 can be assumed to be constants in this example. In a typical real life application, these would be the positive and negative terminals of a rectified mains supply. The T_1 and T_2 transistors control the entire circuit.

Let us consider first the simplest and basic solid-state DC-AC inverter, namely the single-phase half-bridge inverter show in Figure 3.6 It consists of two switching devices (T_1 and T_2) with two antiparallel diodes (D_1 and D_2) to accommodate the return of the current to the DC bus when required. This happens when the load power factor is other than unity. In order to generate a mid-point to connect the return path of the load, The result is that the voltage V_{dc} is split into two equal sources across each capacitor with voltage of $E_a/2$. The assumption here is that the value of the capacitors is sufficiently large to ensure a stiff DC voltage source. This simply means that their voltage potential remains unchanged during the operation of the circuit. This also means that the potential of the mid-point is constant with respect to both positive and

negative DC bus rails at all times ($E_a/2$) and ($-E_a/2$) respectively shown in figure (3.6).

The inverter discussed here operates in all four quadrants of output voltage and current. There are two distinct modes of operation associated with the transfer of power from the DC to the AC side. When the power flows from the DC bus to the AC side, the converter operates as an inverter. The switches T1 and T2 perform this function. In the case that the power is negative, which means power is returned back to the DC bus from the AC side, the converter operates as a rectifier. The diodes D1 and D1 perform this function.

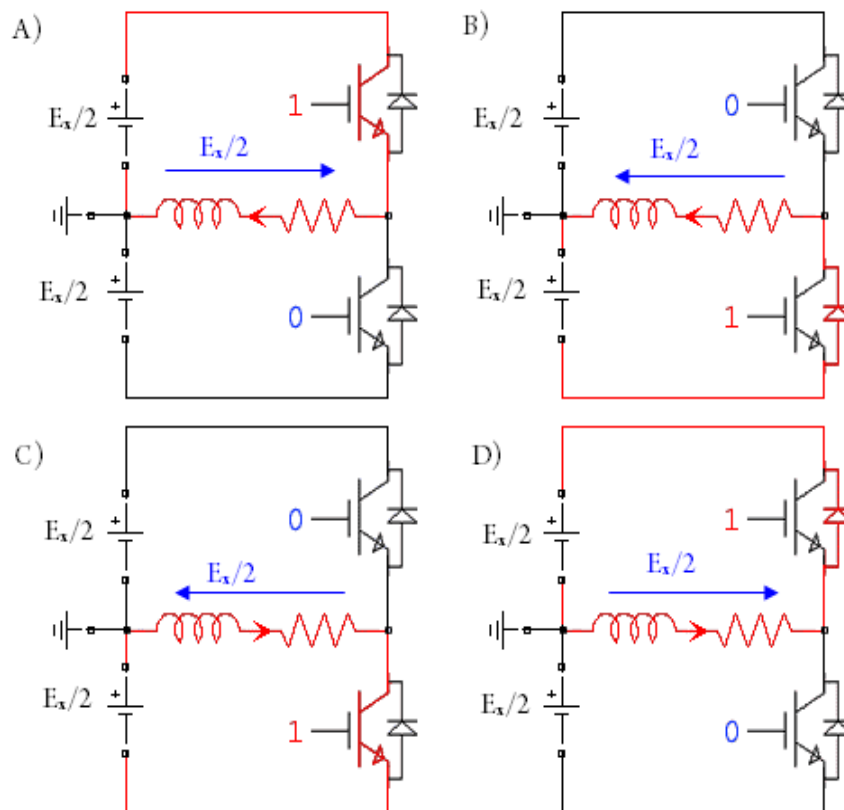


Fig.3.6. Four phases of operation of a half-bridge inverter

During phase (A), T_1 is conducting – therefore a positive voltage appears across the load and there's a current through the load. This current is slowly getting larger. During phase (B), the polarity of the load voltage is flipped (inverted). Therefore, the current of the inductor has “built up” then starts decreasing and

reversing polarity. Notice that during this stage, the current of the load is **opposing** the voltage as if it was a **current source**, and the load current flows through T_2 's diode, flowing against the barrier of potential created by the bottom voltage source show in figure(3.7).

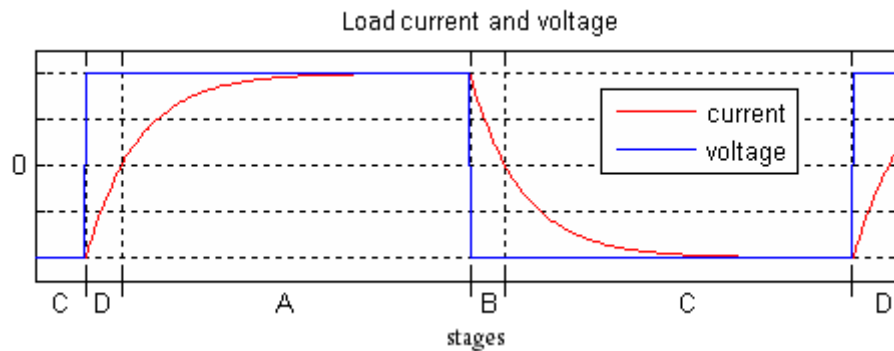


Fig.3.7. Voltage and current during the four stages of the operation of a half-bridge (simplified)

During phase (C), the load current has already changed its polarity, so it can safely flow through T_2 . This phase is symmetrical to phase (A). Phase (D) is analogous to phase (B), the current cannot find any escape path other than T_1 's diode.

This should suffice to explain why are fly back diodes used with inductive loads. Since an IGBT can conduct current in one direction only, and does not have an intrinsic body diode of a MOSFET, this is why IGBTs need to have protective freewheeling diodes in parallel.

Using a diode in parallel with a unidirectional device is common practice. Should the device be subjected to a reverse-polarity voltage, a dangerous situation might occur. If the voltage is high enough, an electric arc might destroy the device. With a diode, however, applying a reverse voltage will result in current flowing through the diode. This may only destroy the diode, which can be easily and inexpensively replaced.

3.5.3 Full H-Bridge Single Phase Inverter

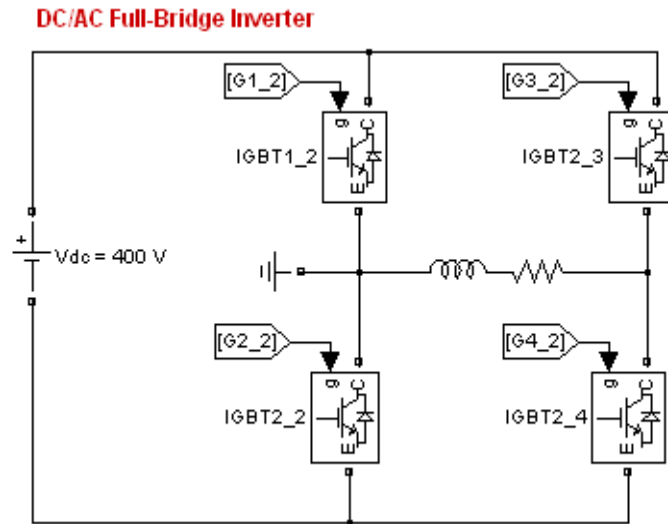


Fig. 3.8. H-Bridge inverter

In this section we will examine in detail the single-phase full-bridge inverter circuit is shown in figure (3.8). It consists of two identical legs like the half-bridge single-phase inverter figure (3.6) discussed. Specifically, there are four switching elements (T_1, T_2, T_3, T_4), four antiparallel diodes (D_1, D_2, D_3, D_4) and a DC bus voltage source V_{dc} that can be a single capacitor. The other leg provides the return path for the current this time and the DC bus mid-point does not need to be available to connect the load. The output voltage v_o appears across the two points of the load as shown in Figure (3.8).

The control restriction discussed for the single-phase half-bridge topology (Figure (3.6)) applies to this converter as well. Clearly the control signals for the switch pairs (T_1, T_2) and (T_3, T_4) must be complementary to avoid any bridge destruction due to shoot through of infinite current (at least theoretically).

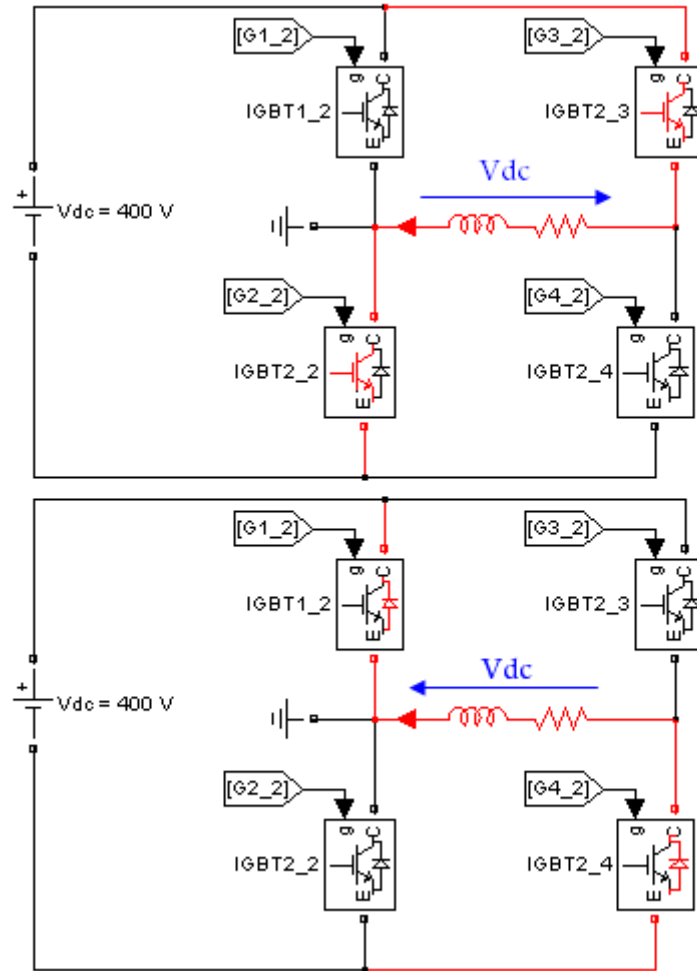


Fig.3.9. The two main phases of operation of a H-Bridge

There are two control methods for this topology. The first one treats the switches (T1, T4) and (T2, T3) as a pair. This means that they are turned on and off at the same time and for the same duration. For square-wave operation the switches T1 and T4 are on for half of the period. For the other half, the pair of T2, T3 is turned on. Like the single-phase half-bridge inverter, the direction of the output current i_o determines the conduction state of each semiconductor. When the two switches S1 and T4 are turned on, the voltage at the output is equal to the DC bus voltage V_{dc} . Similarly, when the switches T2 and T3 are turned on the output voltage is equal to $-V_{dc}$.

In the first case, when the direction of the output current i_o is positive as shown in Figure 3.6, the current flows through switches T1 and T4 and the power is

transferred from the DC side to the AC one ($t_4 < t < t_5$). When the current becomes negative, although the switches T1 and T4 are turned on, the diodes D1 and D4 conduct the current and return power back to the DC bus from the AC side ($t_3 < t < t_4$). For the other half of the period, when the switches T2 and T3 are turned on and the current is positive, the diodes D2 and D3 conduct ($t_1 < t < t_2$). In this

3.6 Sinusoidal Pulse-Width Modulation

The Pulse Width Modulation (PWM) technique is applied in the inverter (DC/AC converter) to output an AC waveform with variable voltage and variable frequency. The implementation of the complex PWM algorithms have been made much easier due to the advent of fast digital signal processors, microcontrollers, and Field Programmable Gate Arrays. The principle is simple in Fig (3.10,a,b):

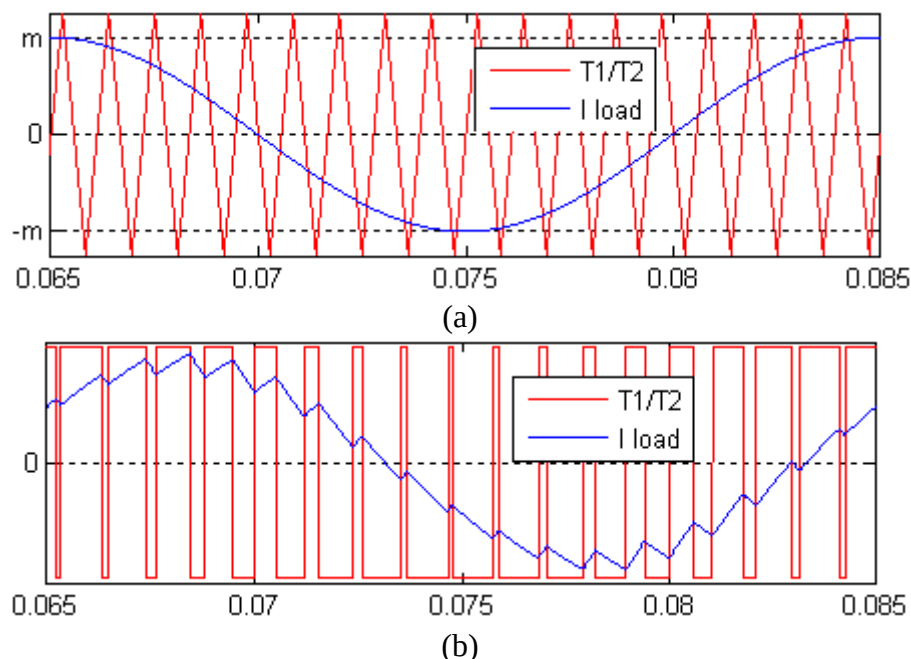


Fig.(3.10.a,b) How does SPWM work

In PWM, cooperation of a desired sine wave with a triangular waveform is done. For the half-bridge, the polarity of the T_1/T_2 switching devices will be positive whenever the desired waveform is bigger than the triangular waveform at any given moment of time. The polarity is reversed when the triangle is larger

than the sine. This sort of control can be implemented using a mere bunch of comparators and op amps, no microprocessors required! It's cheap, easy to debug, troubleshoot, and repair.

Full bridge, PWM modulation is slightly different and a little bit more complicated to implement using discrete logic. The output is set to $+V_{DC}$ when the desired positive voltage is higher than the modulation voltage. When the positive voltage is lower, the output is set to 0V. For a negative input voltage, the output is set to $-V_{DC}$ when $V_{IN} < V_{TRI}$, and 0V otherwise.

Increasing the frequency of the triangular waveform will increase the switching frequency of the bridge. This improves the accuracy of modulation, lowers noise and distortion, but has its drawbacks. It pushes the limits of IGBTs, which are a bottleneck of this circuit. IGBTs are getting faster as manufacturers improve their designs, but at the present time, 20kHz can be considered an upper limit for them. For higher frequencies, you'd want to use MOSFETs – but these come with their own problems.

The desired waveform has two major properties: frequency and magnitude. The frequency should be limited – if it is larger than the triangular wave frequency, the output will be aliased. The maximum theoretical frequency of the output is half of the frequency of the triangular wave. The resultant waveform, however, would be hardly a sinusoidal one. It is then desirable to limit the sinusoidal wave to one order of magnitude less than the modulation frequency.

The magnitude of the desired waveform is also limited. Full modulation will only occur if the maximum amplitude of the waveform does not exceed the amplitude of the modulation waveform. defined as $m = M_{ref} / M_{carrier}$.

This ratio is used to describe the input magnitude, when m should be higher than one, full modulation will no longer occur. the output waveform would resemble a square wave impeded by an RL load.

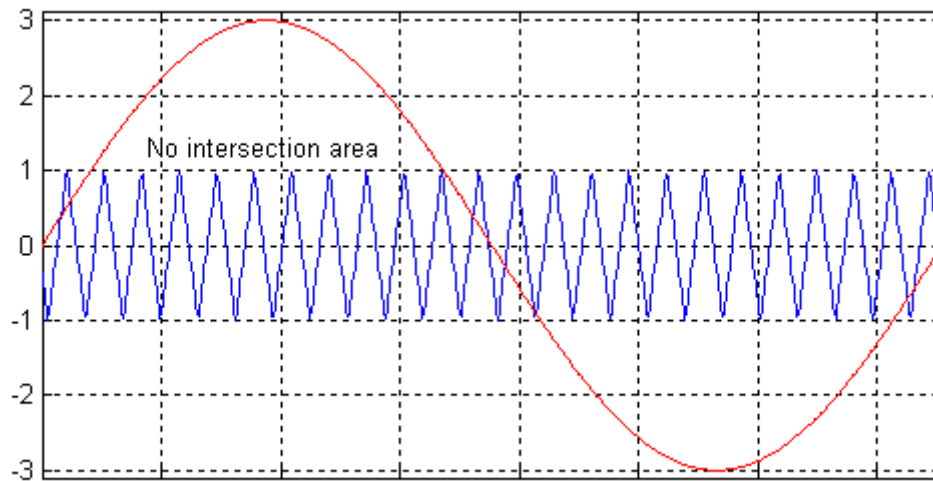


Fig.3.11. Reference and carrier waves.

It's important to know how a different topology can change the performance. The figure(3.12)below show 30% output of a full bridge compared with a half-bridge.

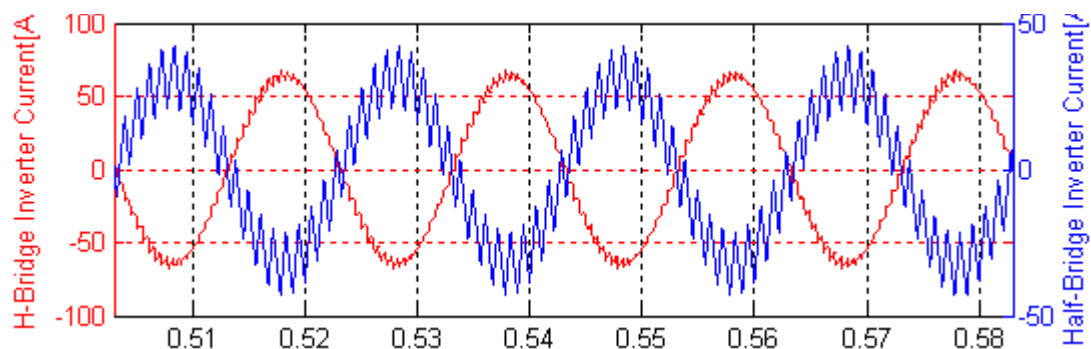


Fig.3.12. Ideal output of SPWM modulation at $m=0.3$ of a full- and half-bridge

Note that this is only an idealized simulation. In reality, each one of these spikes would cause additional ringing and, in some cases, complete instability due to resonance. It should be then clear that the additional resolution of modulation makes quite a big difference. In order to quantify exactly how much of a

difference, some kind of measure of distortion introduced. One such measure is total harmonic distortion (THD). It's defined as:

$$THD = \frac{\sqrt{\sum_{h=2}^{\infty} (M_H)^2}}{M_{fundamental}} \cdot 100 \quad (3.10)$$

$M_{fundamental}$ is the magnitude of the fundamental frequency of the output. M_h in the numerator corresponds to a magnitude of a h^{th} harmonic of the fundamental frequency. This ratio tells us how much energy is wasted in high harmonics. Usually, all harmonics except for the fundamental frequency don't contribute anything useful.

figures (3.13) below shows the THD and the fundamental magnitude of ideal modulated outputs of a full-bridge. While analyzing the performance of the two inverters, Full-bridge inverter shows better performance in terms of THD. The full bridge does exhibit a small non-linearity within the 0÷1 range.

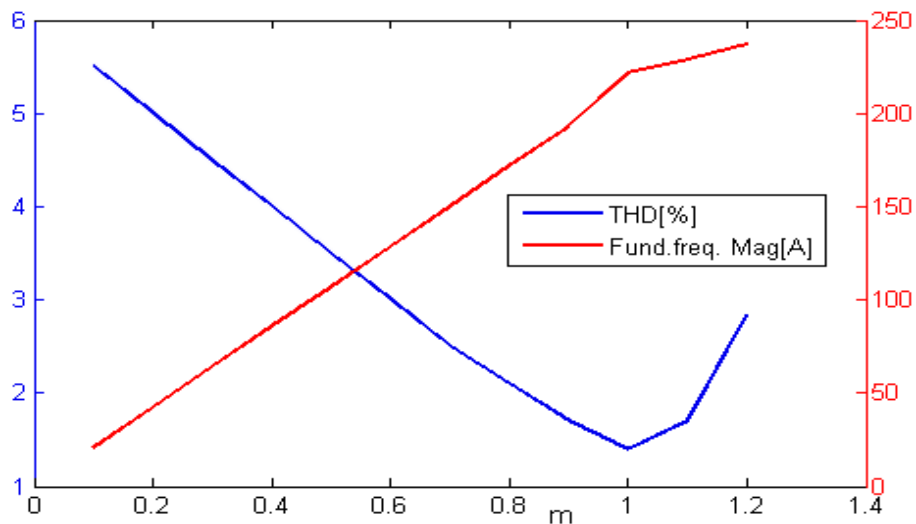


Fig.3.13. Full-bridge performance: THD and output current

3.7 Topologies of Grid Connected PV Systems

In PV plants applications, various technological concepts are used for connecting the PV array to the utility grid[11]. Most inverters generate in one way or another pulse width modulated (PWM) signal, which is filtered to finally

produce a low-distortion sinusoidal signals (voltage and current). The general types of PWM inverter are: Sinusoidal PWM, Harmonic elimination principle, Hysteresis (Adaptive current control PWM), and Phase-shift PWM. Moreover, because the desired power level plays a significant role in switching type selection, therefore for grid-connected PV domestic load systems, MOSFET switches are usually chosen, where they are suitable for low power, high frequency applications. With range of tens amperes and hundreds of voltages.

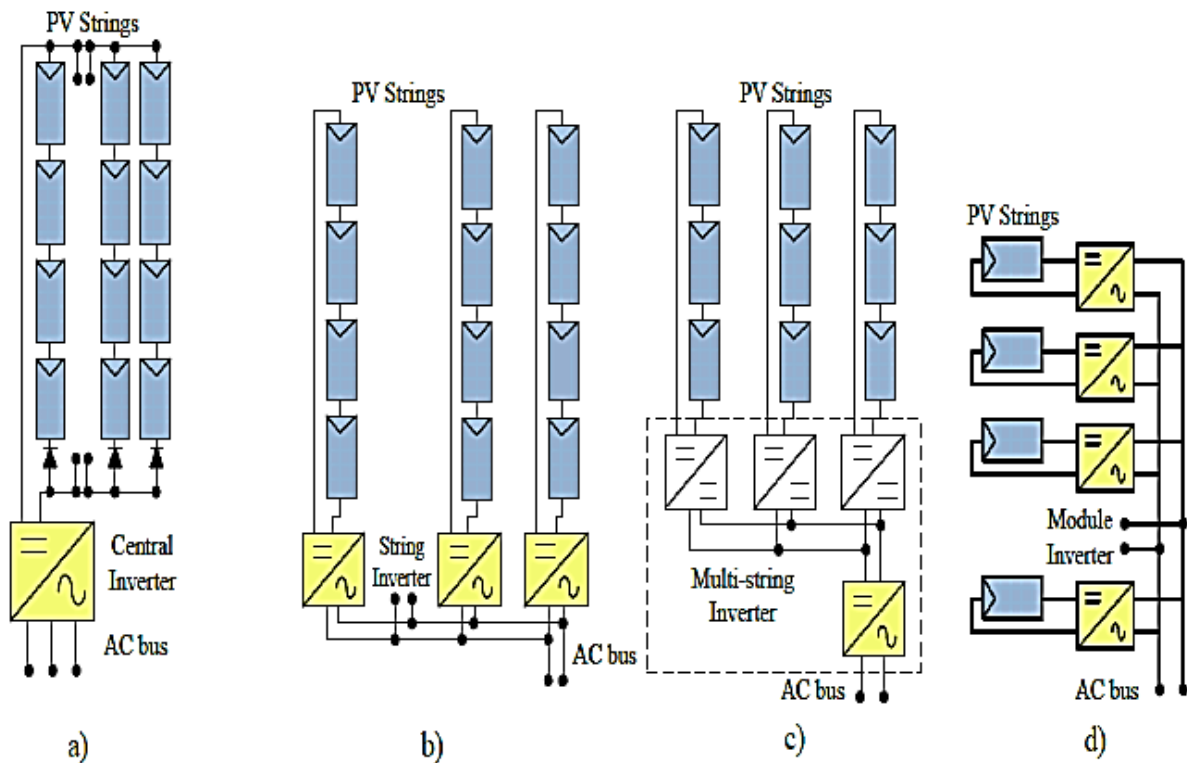


Fig.3.14: PV grid connected systems configurations (a).Central Inverters; (b). String Inverters;
(c).Multi-String Inverters; (d). Module inverters.

The way solar modules are connected to the inverter allows for PV Inverters to be split into five (sometimes four, as mini- and central inverters are sometimes considered together) groups:

- **Module integrated inverters:** typically in the 50-400W range for very small PV plants (one panel) .
- **String inverters,** typically in the 0.4-2 kW range for small roof-top plants with panels connected in one string

- **Multistring inverters**, typically in the 1.5-6 kW range for medium large roof-top plants with panels configured in one to two strings
- **Mini central inverters**, typically > 6kW with three-phase topology and modular design for larger roof-tops or smaller power plants in the range of 100 kW and typical unit sizes of 6,8,10 and 15 kW
- **Central inverters**, typically in the 100-1000 kW range. They also have a three-phase topology and modular design and are intended to be used for large power plants ranging to tenths of a MW and typical unit sizes of 100, 150, 250, 500 and 1000 Kw.

3.8 Standards On Grid-Connected PV Systems

There are multiple standards dealing with the whole topic of PV Energy systems,

ranging from documents about Measuring principles, Solar modules, Components, Power converters, Stand-alone systems and so on.

This standards are often written at an International (IEC) or European (EN) level

and are re-issued at a national level by each's country responsible normative agency (DIN for instance in Germany) Some important standards when considering grid-connected PV inverters are given below, they are referenced to the biggest agency responsible for their development, no matter if later re-issued at a lower scale[8].

Table .3.1: Most important standards on grid-connected PV systems

IEC 60364-7-712	Electrical installations of buildings - Part 7-712: Requirements for special installations or locations - Solar photovoltaic (PV) power supply systems.
IEC 61727	Photovoltaic (PV) systems - Characteristics of the utility interface.
IEC 61683	Photovoltaic systems - Power conditioners Procedure for measuring efficiency.
IEC 62093	Balance-of-system components for photovoltaic systems - Design qualification natural environments.
IEC 62116	Test procedure of islanding prevention measures for utility interconnected photovoltaic inverters.
IEC 62446	Grid connected photovoltaic systems – Minimum requirements for system documentation, commissioning tests and inspection.
IEC 62109	Safety of power converters for use in photovoltaic power systems – Part 1: General requirements
EN 50530	Overall efficiency of grid connected photovoltaic inverters
EN 50160	Voltage characteristics of electricity supplied by public electricity networks
IEEE 1547	Interconnecting distributed generation resources with energy power systems

3.9 Standards Focusing on Current and Voltage Harmonics

3.9.1 EN 50160

Maintaining a satisfactory PQ is a joint responsibility of the supplier and the electricity user. The supplier is the party that provides electricity via a public distribution system and the user or consumer is the party that purchases electricity from the supplier. The user is entitled to receive a suitable quality of power from the supplier. In practice the level of PQ is a compromise between user and supplier. EN 50160 standard, issued in Germany by DIN, is the main document dealing with requirements concerning the supplier's side. This standard characteristic voltage parameters of electrical energy in the public distribution systems. The European standard is supplemented in some specific countries. For the case of Germany, there are also the so-called (Technical connection requirements) The main voltage parameters and their permissive deviation ranges at the customer's point of common coupling (PCC) for low (LV), medium (MV) and high voltage distribution systems under normal operating conditions is given in EN 50160. It has to be noted, that this limits are only general, and economically reachable for the supplier. When more specific limits are to be reached, an special agreement between user and supplier should be met. Among other properties of the supply voltage, harmonics are also considered in this standard. Limit values for the RMS voltage of harmonics up to $h=25$ are given.

Table .3.2: Voltage harmonic limits contained in EN 50160

ODD HARMONICS		EVEN HARMONICS	
Not multiple of 3	Multiple of three	Order h	Relative

Order h	Relative voltage (%)	Order h	Relative voltage (%)	voltage (%)	
5	6.0	3	5	2	2.0
7	5.0	9	1.5	4	1.0
11	3.5	15	0.5	6 to 24	0.5
13	3.0	25	0.5		
17	2				
19	1.5				
23	1.5				
25	1.5				

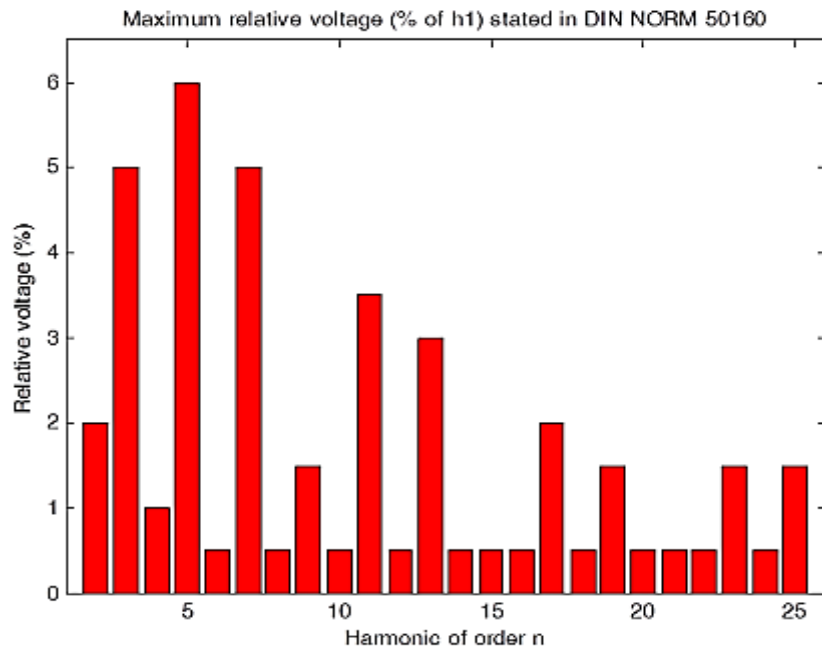


Fig.3.15: Maximum voltage harmonics contained in EN 50160

3.9.2 IEEE 1547

The IEEE standard 1547 (Interconnection of distributed generation) focuses on the technical specifications for, and the testing of, the interconnection of distributed generation itself. It includes general requirements, response to abnormal conditions, power quality, islanding and test specifications as well as requirements for design, production, installation evaluation, commissioning and periodic tests.

A huge effort in harmonizing the grid requirements has been done by the IEC that promotes international cooperation within standardizations for electrical and electronic issues. The standard IEC 61727 (PV Systems Characteristics of the Utility Interface) was published in 2004. The standard applies to utility-

interconnected PV power systems operating in parallel with the utility and using inverters for the DC/AC conversion and lays down requirements for the interconnection of PV systems to the utility distribution system. IEC 61727 and IEEE 1547 harmonize very well in some aspects. For instance, the values of relative maximum current harmonics are the same in both standards and shall be considered.

Table .3.3: Maximum current harmonics contained in IEEE 1547

Harmonic number (h)	$h < 11$	$11 \leq h < 17$	$17 \leq h < 23$	$23 \leq h < 35$	$35 \leq h$
Relative current (%)	4.0	2.0	1.5	0.6	0.3

Even harmonics are limited to a 25% of the odd harmonic limits above

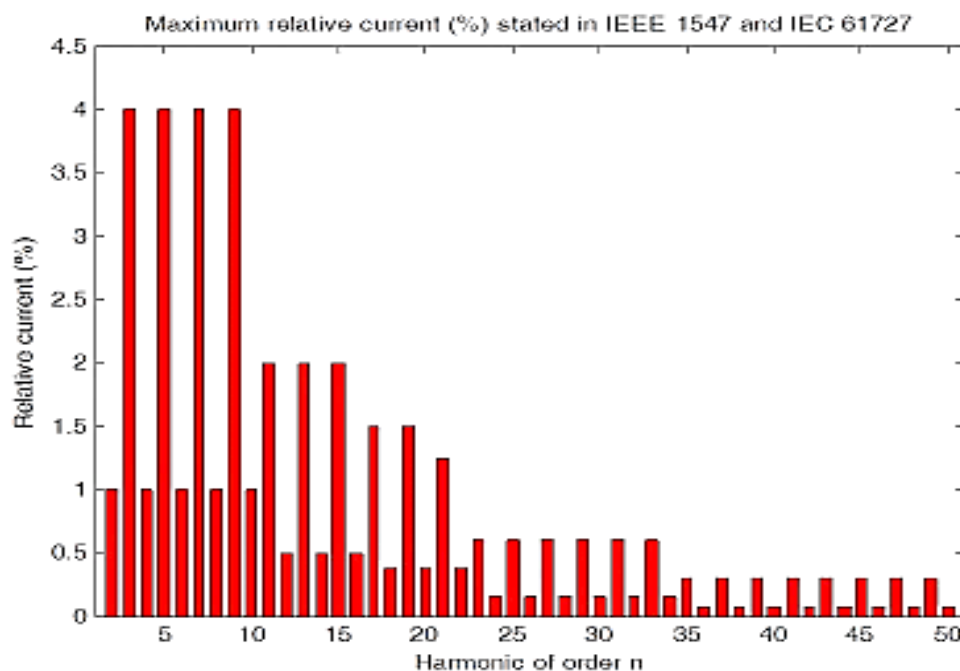


Fig.3.16: Maximum current harmonics contained in IEEE 1547

3.9.3 IEC 61000-3-2 and IEC 61000-3-12

The aim of EMC (Electromagnetic compatibility) is to ensure reliability and safety of all types of systems wherever they are used and exposed to electromagnetic environments. IEC (International Electro technic Commission),

with the international scope of its standards, has developed a series of documents on the field of EMC. The IEC 61000 series reflects the subjects dealing with basic EMC publications. It includes terminology, publications, descriptions of electromagnetic phenomena and the EM environment, measurement and testing techniques and guidelines on installation and mitigation. Specifically, part 3 sets limits for emission and immunity. Particularly, IEC 61000-3-2 addresses and sets limits for equipment that draws input currents ≤ 16 A per phase. IEC 61000-3-12 focuses on equipment with input currents between 16 and 75 A.

Table .3.4:Maximum current harmonics contained in EN-61000-3-2

<i>Harmonic number (h)</i>	<i>Maximum absolute current (A)</i>
2	1.08
3	2.30
4	0.43
5	1.14
6	0.3
7	0.77
9	0.40
11	0.33
13	0.21
15≤h≤39 (Odd harmonics)	0,15*15/h
8≤h≤40 (Even harmonics)	0,23*8/h
All even harmonics lower than h=42	1,5/h
42<h<178 (Even and odd harmonics)	4,5/h

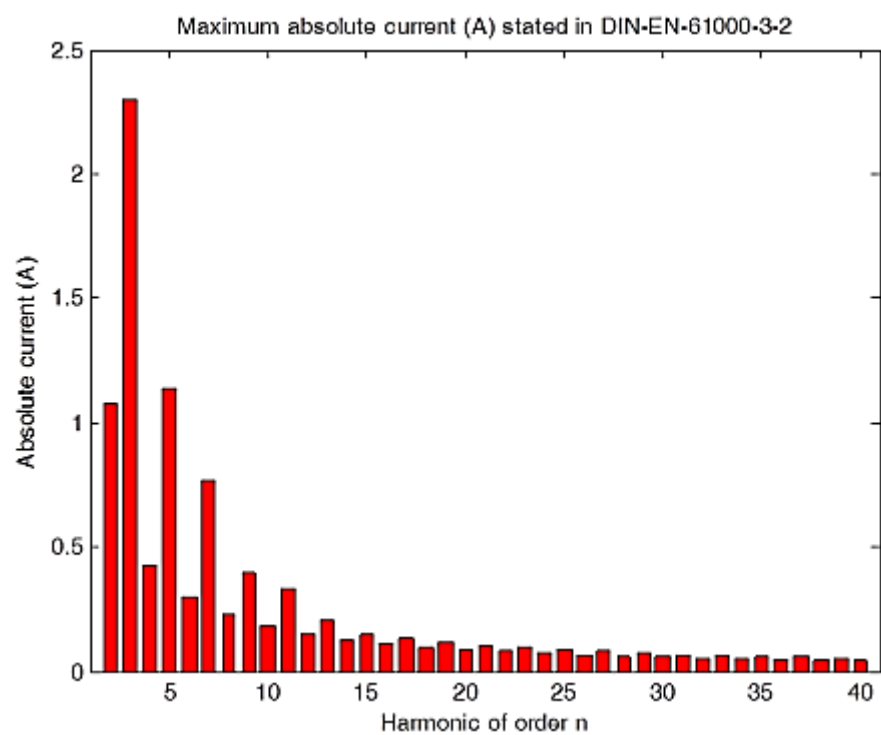


Fig.3.17: Maximum current harmonics contained in EN-61000-3-2

CHAPTER FOUR

POWER QUALITY FIELD MEASUREMENTS OF PV INVERTERS

4.1. General

In this chapter various measurements of grid connected inverters of small scale PV stations have been carried out during real operating conditions. Then, a comparative study on main electrical characteristics is performed, utilizing the measurement results.

4.2 Grid Connected Photovoltaic System Description

An experimental observation study of grid-connected photovoltaic (PV) system that is installed at Tripoli university, Libya (Latitude 30.04°N , Longitude 31.21°E), is presented. A 24 kW peak (kWp) grid-connected PV power generation system, it is comprised of 240 Mitsubishi a-Si PV modules MA100T2 having a total area of 382.77m^2 . The Modules were constructed on a plane angle equal to 45° as shown in Fig.4.1.



Fig.4.1: shows the array field

The power quality parameters at the inverter output side have been measured using Fluk 435 power quality analyzer. The produced power by the system is injected directly into the grid without storage device. The purpose of this chapter is to present and evaluate the measurements of the power quality parameters obtained from the PV site. Also, this chapter presents a comprehensive evaluation of the performance of the system over a period of one

week. The observation and analyses exploitation of the collected data can help to evaluate the performance of the PV system connected to the network.

The field array is divided to five sub arrays each consist of 48 modules, DC to AC power inverter fabricated by SMA company of the type (SMA Sunny Mini Central) are used for conveying the AC power generated to the utility grid. The data measured by the station's sensors is monitored and transferred to the computer system through SMA Sunny Webbox, then these measurements are classified and

evaluated to determine the station's performance and other important indicators which will help to know the feasibility of utilizing this type of PV generating stations on the southern shore of the Mediterranean.

4.3 Power Quality Field Measurements of PV Inverters

In order to evaluate the harmonic content of the grid current at the output of the PV inverter, field tests have been performed under real operating conditions. The power analyzer used for all the tests is a FLUKE 435type, certified as Class A according to international IEC 61000-4-30 standard. The Power analyzer was connected to the output of the inverter. Four amp clamps measure the current of each phase and neutral too, while 5 wires were connected to each phase, on neutral and ground in order to measure the voltage.

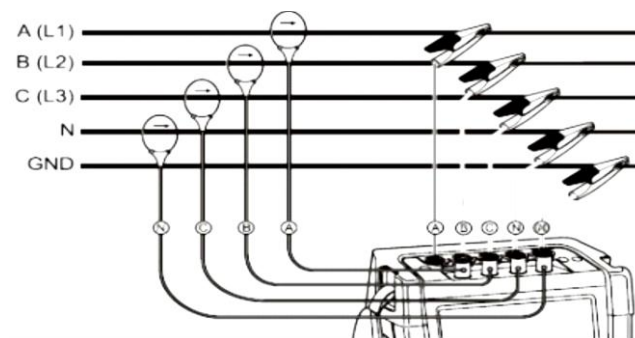


Fig.4.2: Test set up

4.4 PV System Layout

Fig. (1) shows the layout of the PV system connected to the grid. The rated or nominal power of the PV system is $P_n = 24$ kW. which have been oriented towards South. The PV modules are connected in series, forming the PV arrays that are in turn connected in parallel to obtain a PV unit with appropriate nominal power. The PV arrays are connected to the Low Voltage grid through 5 inverters. Table 1 shows the electrical characteristics of the PV modules, while Table (2) shows the mechanical characteristics.

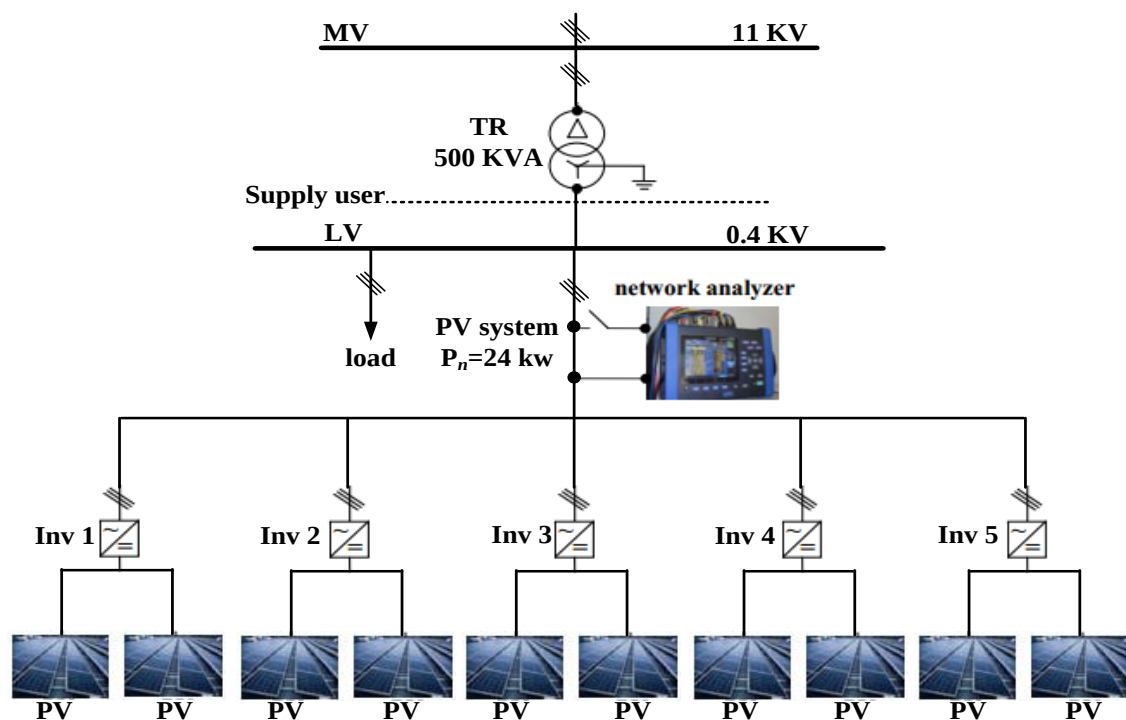


Fig4.3. Layout of the PV system connection to the Medium Voltage grid.

Table.4.1.In Standard Test Conditions (STC): solar irradiance 1000 W/m², air mass (AM) 1.5 and cell temperature 25°C.

Rated power (P_{max})*	100 W
Voltage at the maximum power point (V_{mpp})*	108 V
Current at the maximum power point (I_{mpp})*	0.93 A
Open circuit voltage (V_{oc})	141 V
Short circuit current (I_{sc})	1.17 A
PV module efficiency	96.1 %
Range of operation temperatures	-25 C° ... +60 C°
Temperature coefficients	P_{max} -0.20 % / °C
	V_{max} -0.32 % / °C
	I_{max} +0.14 % / °C
	V_{oc} -0.33 % / °C
	I_{sc} +0.09 % / °C

Table.4.2.Mechanical characteristics of a PV module [1]

Module type	Thin film PV module (MA 100 T2)
Dimensions	L 4,414 × w 1,114 × T 35 mm
Weight	Approx 21 kg

This proposed PV system configuration is important in the design of the power stage of the inverter. The output power rated will be assumed to be 1kW. This value is assumed as the worst case; hence the actual rated power from the string is 24 kWp for ideal PV system. Considering factors affecting power conversion from PV system this power would drop further to a value lower than the ideal rated power.

4.5 Grid-Connected Inverter

This inverter is turned on all the day time and automatically synchronize to the electric grid. If the electric grid has a problem like shut-down or unusual problem, the inverter stops its operation for operator safety [2]. Table 4.3 shows the grid-connected inverter parameters.. This inverter shows on its screen the

PV system power transmitted to the electric grid, system voltage, amount of produced energy during the day and total run time, etc.

Table.4.3. Grid-connected inverter specifications.

DC side (PV generator connection)	
Maximum input voltage (VDCmax)	600 V
Rated input voltage	246 V
AC output side (mains grid connection)	
Rated output voltage	240 V
Rated output current	26 A
Rated output power	5000 W
Max. apparent power ($\cos\phi = 0.90$)	5500 VA
Grid frequency	50 Hz
Power factor $\cos\phi$	1
Maximum efficiency	96.3%

Grid-connected PV systems utilize DC/AC converters which convert the DC current generated by solar panels into AC, in accordance with the electrical characteristics of the grid to which they are connected. The harmonic distortion induced by PV systems at the Point of Common Coupling (PCC) of a Low Voltage (LV) grid should be sufficiently low in order not to deteriorate the quality of supply to customers connected to the same LV grid.

4.6 Measurement Device

In order to evaluate the harmonic content of the grid voltage and current at the output of the PV inverter, field tests have been performed under real operating conditions. The power analyzer used for all the tests is a FLUKE 434 type, certified Class A according to international IEC 61000-4-30 standard. Figure 1 shows the general test set up [12].



Fig.4.4: test set up

The Power analyzer was connected to the output of the inverter. Four Amp clamps measure the current of each phase and neutral too, while 5 wires were connected each phase, on neutral and ground in order to measure the voltage.

Power Log is the PC Software for FLUKE 434. The software processes data acquired by the instrument. Logged data can be transferred to a PC for graphical and tabular evaluation, where they can be exported to a spread sheet for generating and printing reports.

4.7 Methodology of Data Collection

The power analyzer outputs are the following:

- Harmonics measurements for harmonics up to 40th order. This enables the evaluation of the harmonic content of the measured grid voltage and current waveforms.
- Voltage / Current / Frequency measurements. Voltage measurements include both neutral-line and line-line RMS values.
- Real Power / Reactive Power / Apparent Power. A Meter screen displays the power measurements, also There is a Trend screen which displays the power deviations of each power component over time.

4.8 Variability of the Power Generated by the PV System

The measured data were analyzed during the period 2013-2014. Fig.4.5 presents the variation of the active power generated by the PV system with installed

power 24 kW, on the time span of one week. The measurements have been carried out in the location shown in Fig. 1, that is, measuring the entire generation from the PV system at the point of common coupling of the grid-connected inverters. The results show that the active power generation follows the evolution of the meteorological conditions.

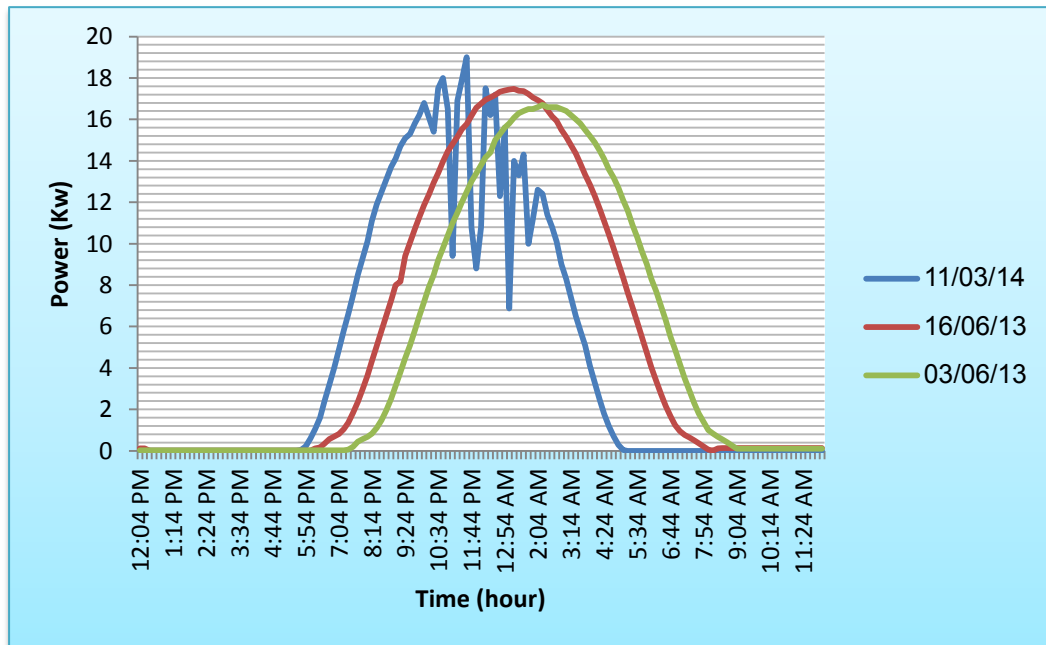


Fig. 4.5. Load generation of the grid for 2013 and 2014

4.9 Power Quality on The Grid Measured at The PCC

Fig. (4.7,4.9,4.11) shows the phase voltages and reports the phase currents produced in the single-phase system during different days of the week. The voltage variations are fully acceptable, as the phase voltage oscillates in the range between 228 V and 217 V (all the values belonging to the normal conditions with RMS voltage between 95% and 105% of the rated voltage), while the current depends on the solar irradiance level, especially during the early morning hours and at sunset.

The reactive power varies not significantly during the day as shown in Fig.4.6, 4.8, 4.10. The active power produced by such a photovoltaic system to peak power is 17 kW in the summer while up to 20 kW during low radiation.

Because of changes in solar radiation, the level of solar radiation has a direct effect on the energy of the plate.

Measurements carried out in the PV plant captured a variety of cases during the observed different two weeks. Daily curves of output power of the PV plant for (a) winter week and (b) summer week are depicted in Figure below During the winter period, daylight lasts from about 7:00 to 17:30, while during the summer period it lasts from about 6:00 to 21:00. Daily curves of output power of the PV plant in the summer week, as shown in Figure. (4.6), indicate a large variety in generation because the Spring week was cloudy and rainy especially at the end of the week as shown in Figure(4.10).The correlation between output power and solar radiation is strong. The measurements showed that even short interruptions of small passing clouds may cause sudden dips in output power.

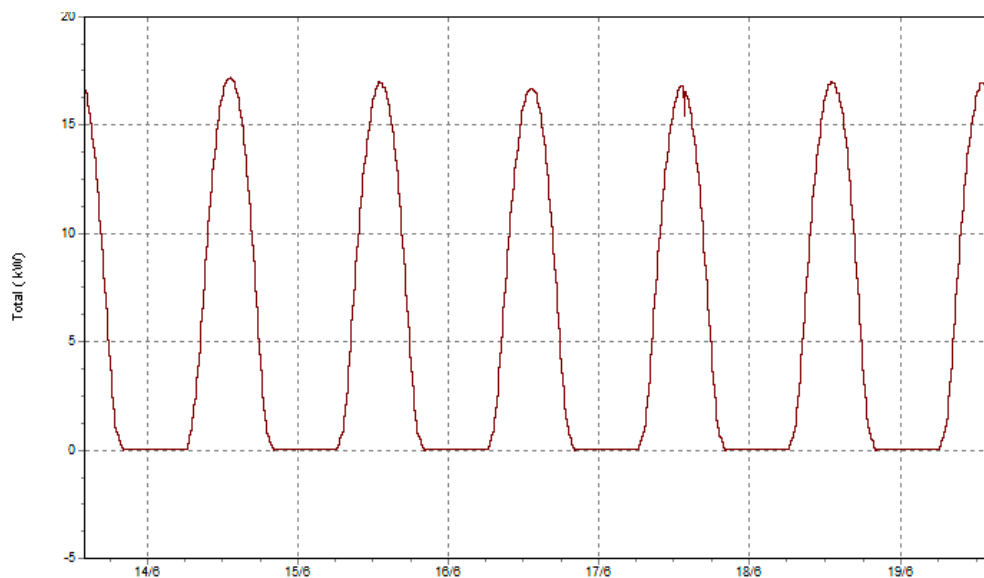
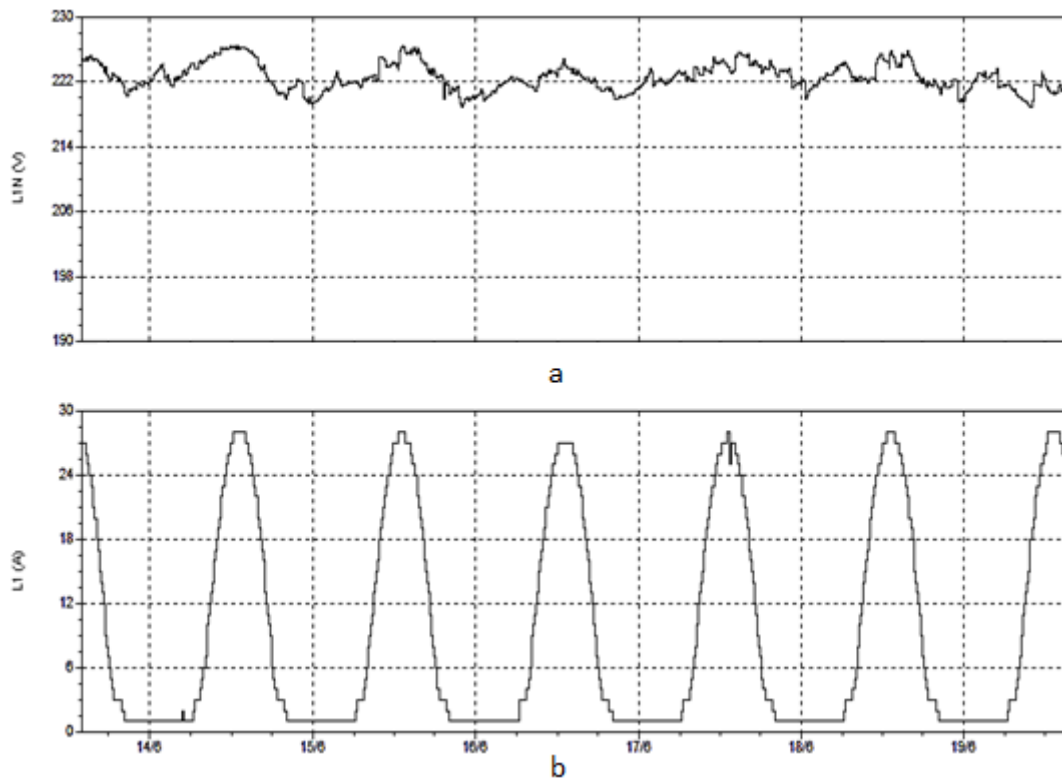


Fig.4.6. Variation of the active power generated of the PV system in one week.

14/6/2013 to 19/6/2013



(a) phase voltage (b) phase currents

Fig. 4.7. Voltage and current at the point of common coupling during the week

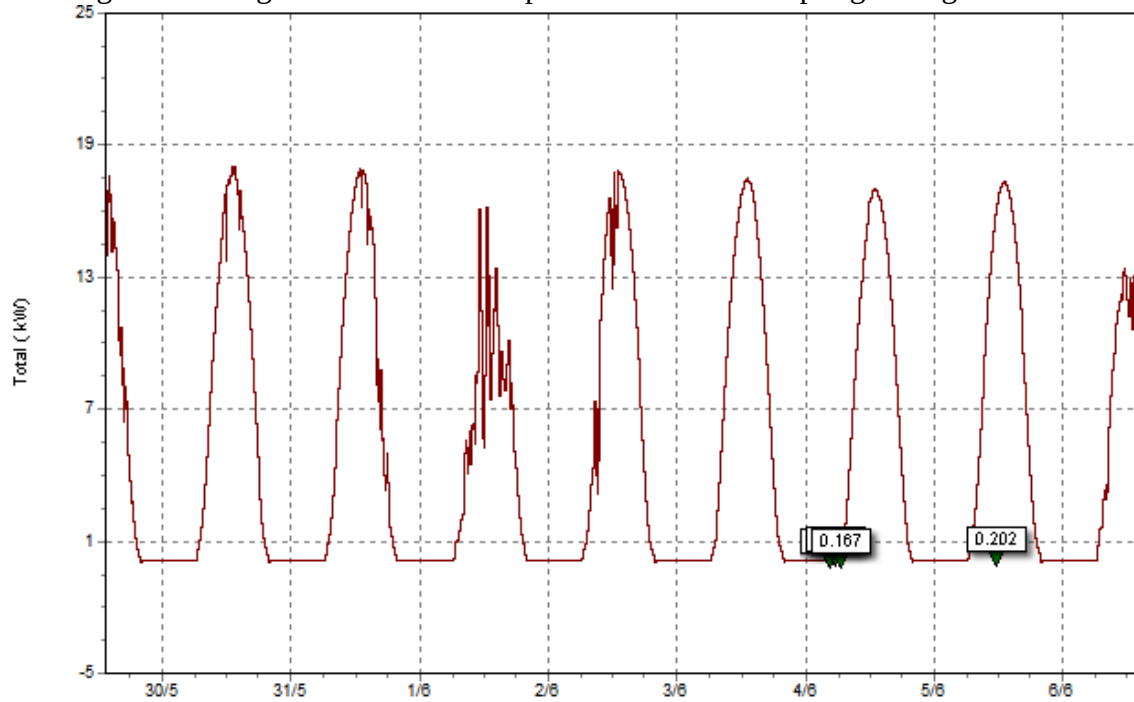
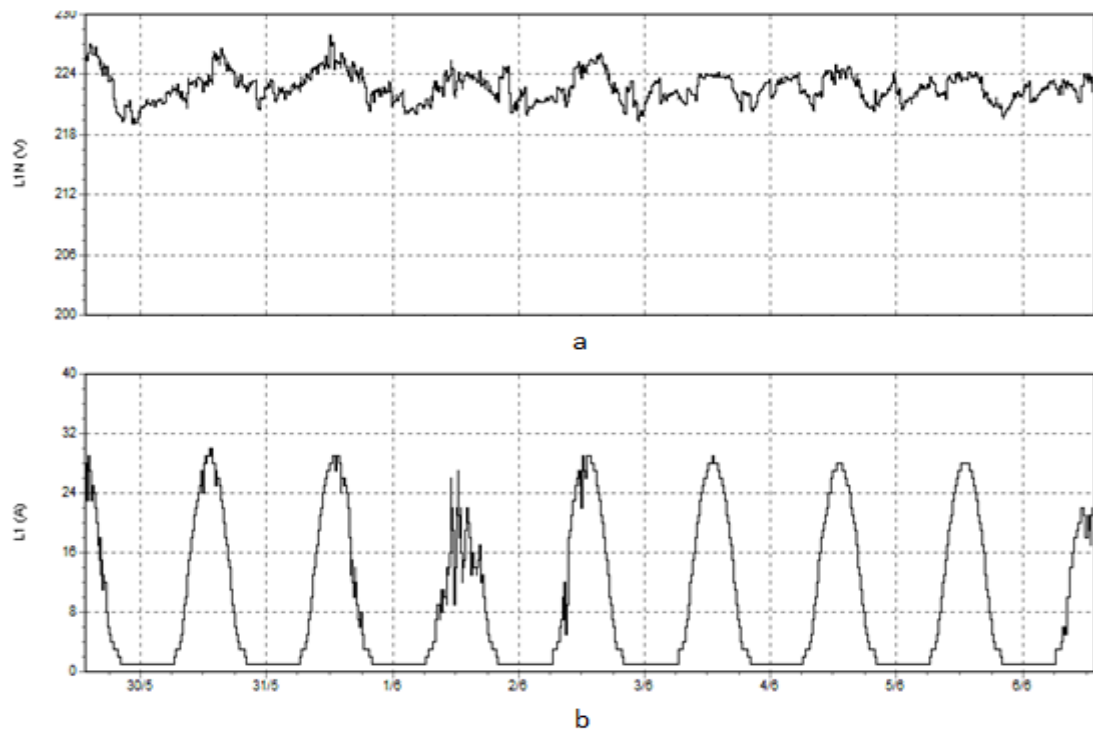


Fig. 4.8. Variation of the active power generated of the PV system in one week.

31/5/2013 to 6/6/2013



(a) phase voltage (b) phase currents

Fig. 4.9. Voltage and current at the point of common coupling during the week

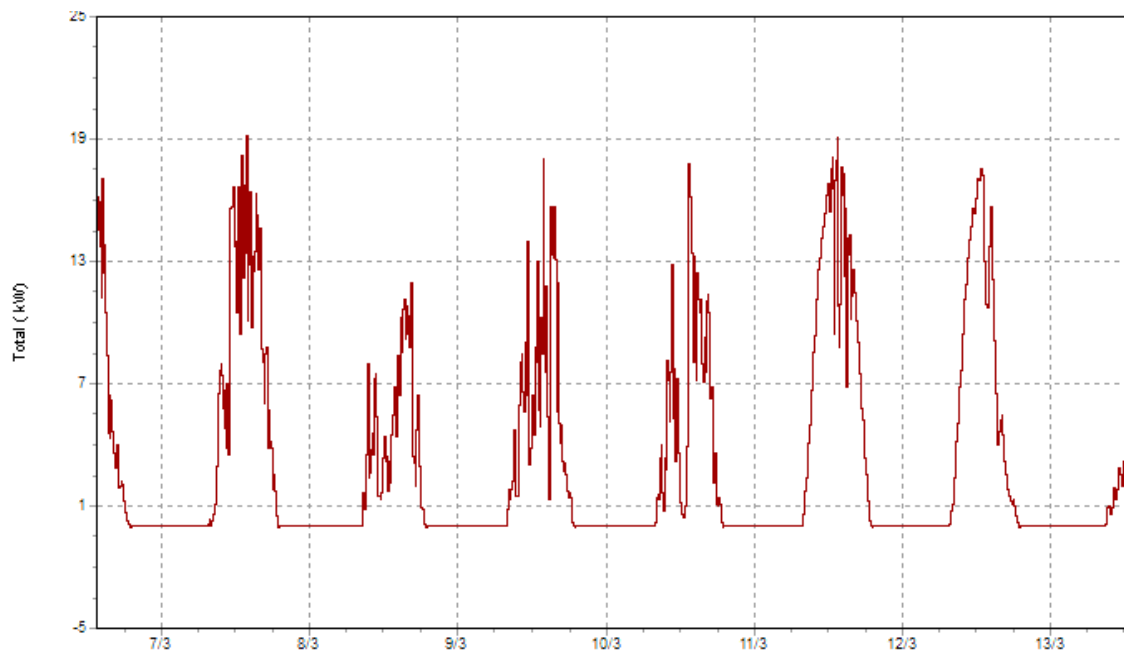
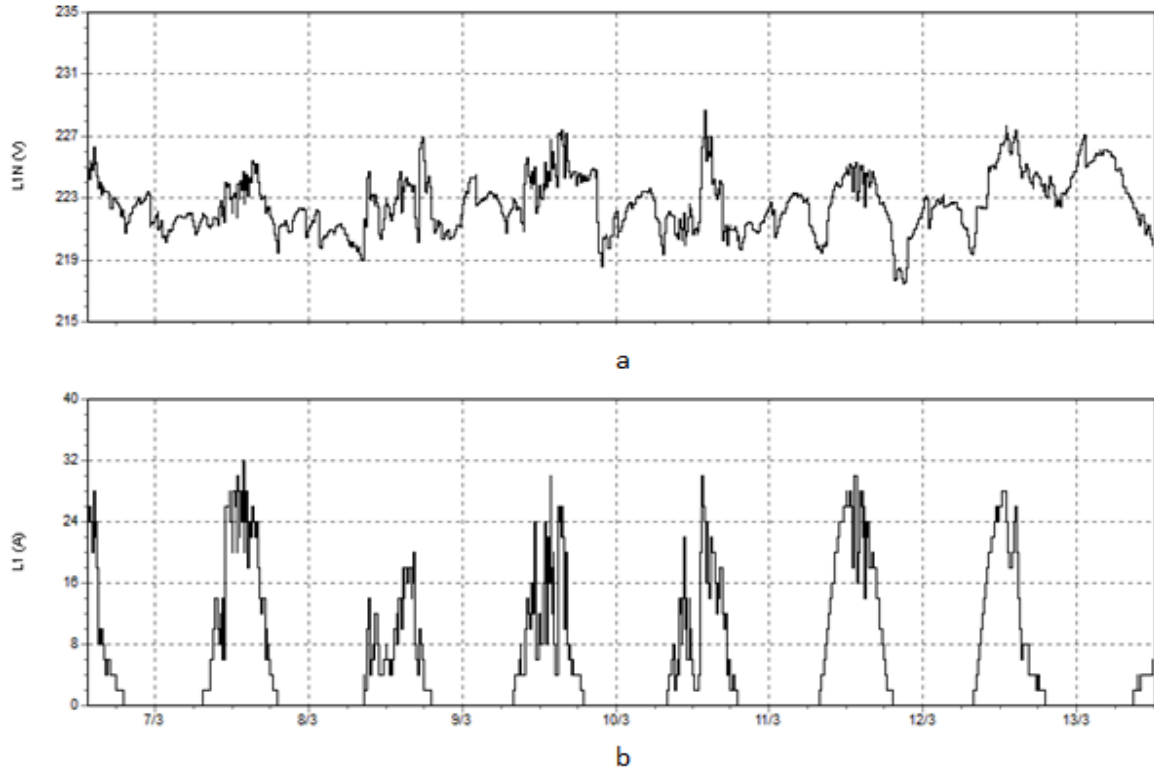


Fig. 4.10. Variation of the active power generated of the PV system in one week.

7/3/2014 to 13/3/2014



(a) phase voltage (b) phase currents

Fig. 4.11. Voltage and current at the point of common coupling during one week

The power factor show in Figure(4.12) is given as a numerical value between 0 and 1 and describes the quantitative ratio of the real power flowing to the load of the apparent power in the circuit. The results for the power factor are found to be acceptable for a large fraction of the day. The AC grid frequency show in Figure(4.13) is the nominal frequency of the oscillations of alternating current in a utility grid transmitted from the solar power inverter to the end user. It can vary in accordance with the corresponding country-specific standards (50 or 60 Hz). The allowable AC frequency range at the output of the solar power inverter is between 49.4 and 50.2 Hz. This parameter is used in the anti-islanding feature of the solar PV inverter. A good solar power inverter has a large AC frequency range.

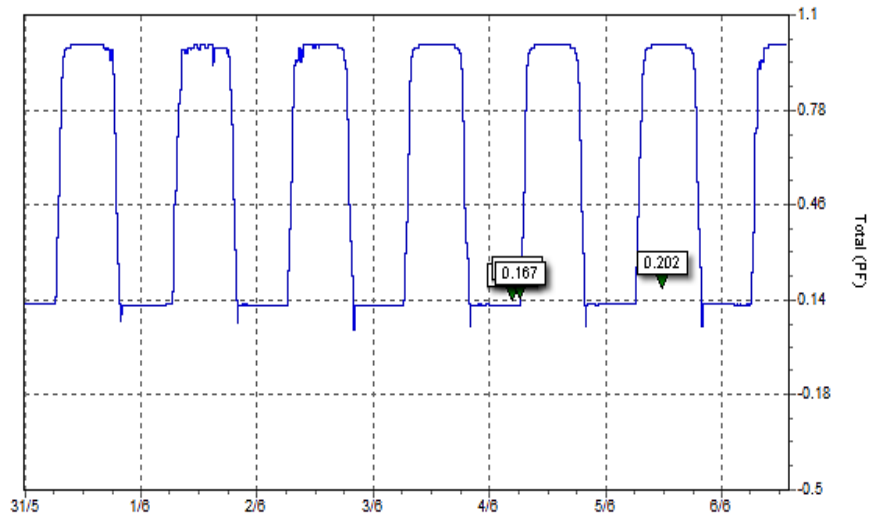


Fig. 4.12 power factor at the point of common coupling during on week
31/5/2013 to 6/6/2013

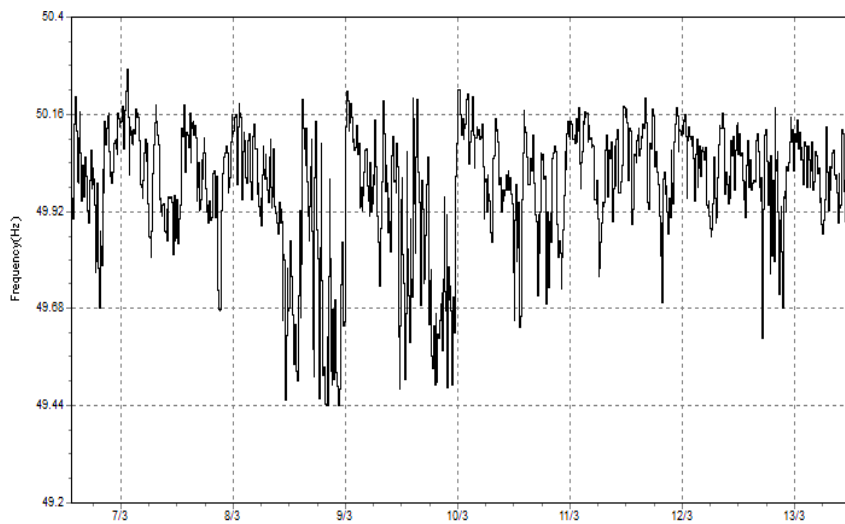


Fig. 4.13 frequency at the point of common coupling during one week

4.10 Assessment of The Current Harmonics

The harmonics referring to the currents produced by the PV system depend on the nature of the source, as well as on the type of inverter technology used for

DC/AC conversion and its control strategy. For the PV system, a particular aspect referring to the current harmonics is the behavior of the inverter in conditions of reduced solar irradiance, also due to the effects of shadowing or passing clouds during the day [1]. The detrimental effects of harmonic distortion can be manifested in many different ways, as the following[16]:

1. Increased heating effect on electrical distribution equipment and cables,
2. Large neutral currents,
3. Malfunction of protective devices like relays,
4. overheating of transformers,
5. Excessive voltage drops,
6. Bus bar overloading,
7. Losses in switching devices, etc

Fig.(4.16) shows the total harmonic distortion THD_i of the phase currents, for different values of the generation level P/P_n . it is possible to conclude that the THD of the phase voltages in the system analyzed remains within acceptable ranges with respect to the limits imposed by the international standards. The THD of the currents becomes relatively high only when the generation level is low. For instance, when the generation level is higher than 20% the THD_i never exceeds 10%, and for generation level higher than 60% the THD_i never exceeds the limit of 5% which is valid for rated power conditions.

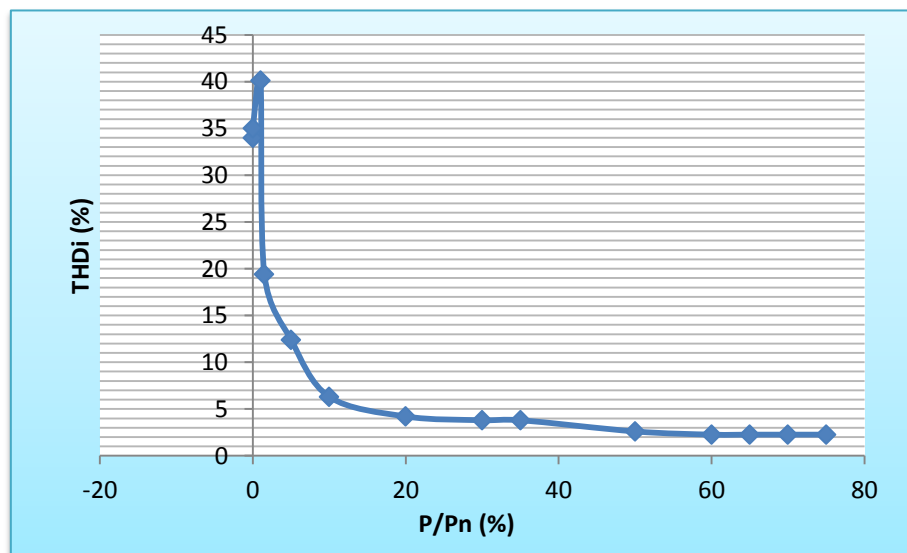


Fig.4.14. Total harmonic distortion of the currents ($THD_i\%$) in one week for different generation conditions.

4.11 Harmonic Currents and Generated Power Measurements

Figures(4.15 to 4.18) illustrates the correlation between the generated power and total current harmonic distortion factor THD_i , for one monitored day (the 7th day of the monitoring period shown in Figure (4.8)). The analysis of the total harmonic distortion factor has to consider that, for high variability installations, the large THD_i values can lead to inappropriate conclusions. The total harmonic factor is related to the fundamental component of the electric current I_1 as follows in equation (4.1):

$$THD_i = \frac{\sqrt{\sum_{h=2}^{\infty} I_h^2}}{I_1} \quad (4.1)$$

For the periods with low irradiation (especially during the morning and evening periods), the electric current injected into the grid presents a reduced fundamental component, resulting in a high distortion factor.

As the electrical current has small values, the voltage drops in the power system are negligible, and thus the voltage waveform at PCC is not affected. When the generated power is high (large value of RMS electrical current), the fundamental current is high and the THD_i is small [16,17].

For small generated powers, the fundamental current is small and the THD_i is high. From a practical point of view, this fact is not highly important as the small current values do not influence the voltage quality at point of common coupling.

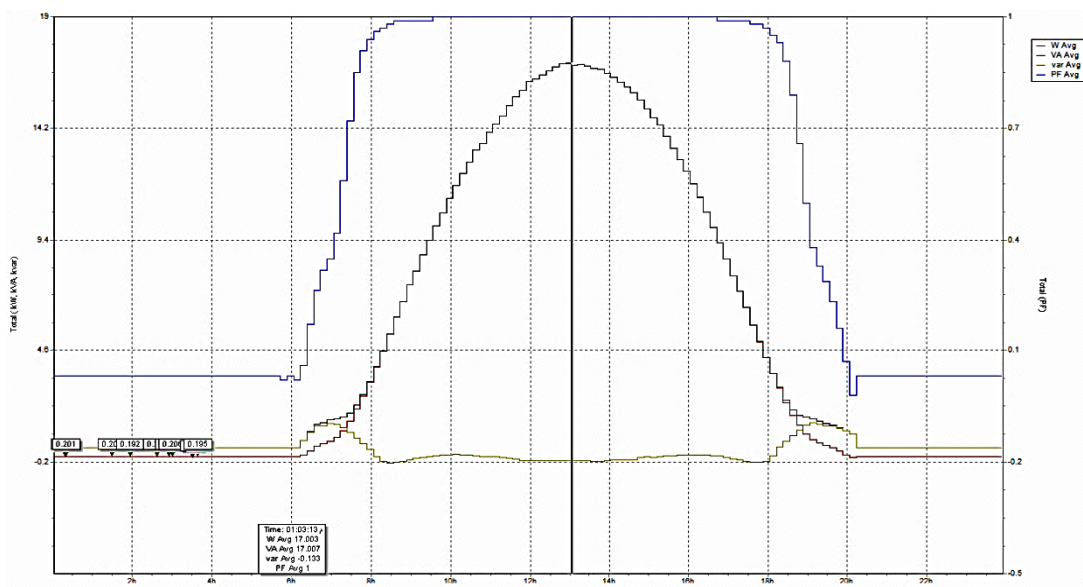


Fig.4.15 Measured active and reactive power and power factor during 15/6/2013
 (— W avg=17.003 Kw , — Var avg=-0.133 , — P.F=1)

Fig.4.16 Measured individual harmonic distortion at early morning to sunset during 15/6/2013

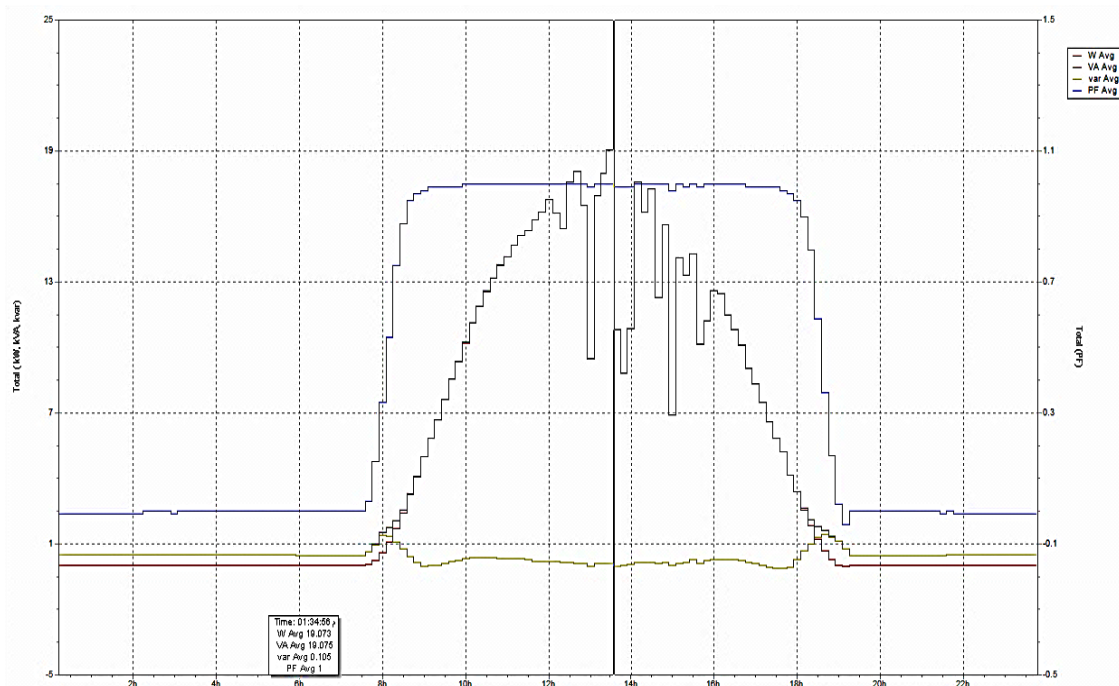
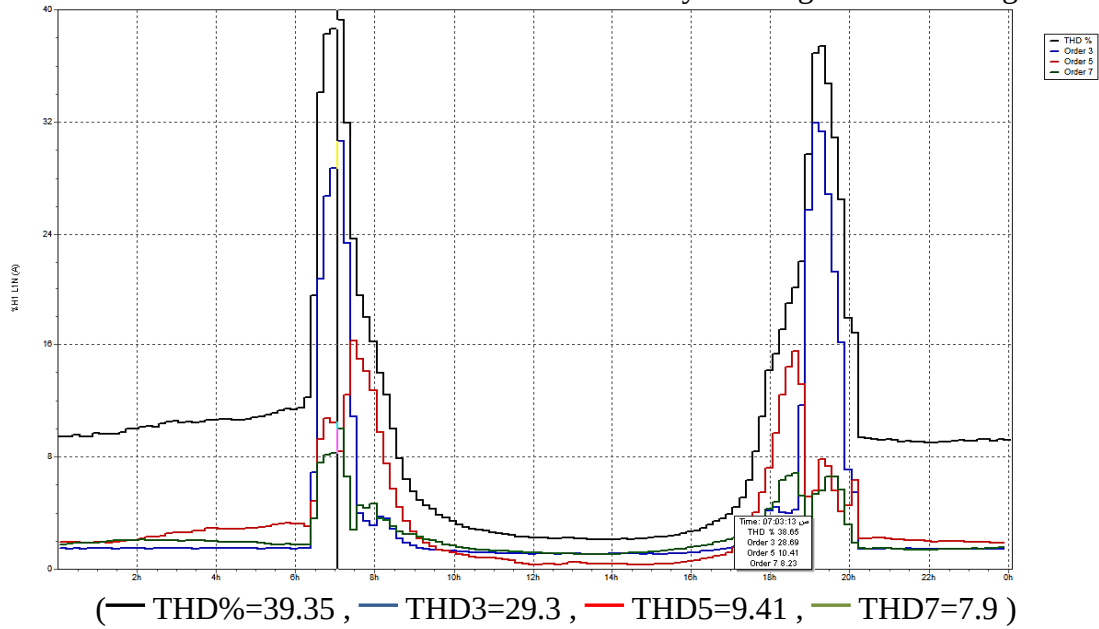


Fig.4.17 Measured active and reactive power and power factor during 11/3/2014
 (— W avg=17.003 Kw , — Var avg=-0.133 , — P.F=1)

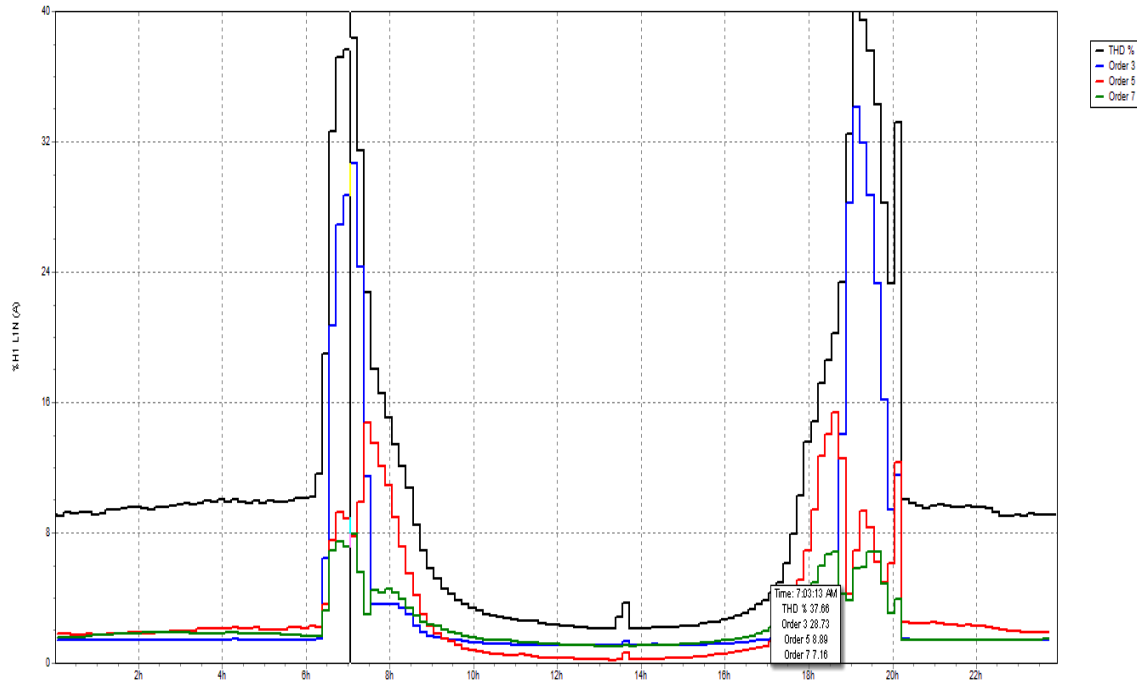


Fig.4.18 Measured individual harmonic distortion at early morning to sunset during 11/3/2014
(— THD%=39.35 , — THD3=29.3 , — THD5=9.41 , — THD7=7.9)

Figure (4.19) illustrates the correlation between the generated power and total current harmonic distortion factor THD_i , for one monitored day (the 5th day of the monitoring period shown in Figure(4.10)). The time evolution of the current THD_i and plant generation during a day with large PV output power variations. As can be seen, the THD_i is characterized by a comparatively high value of up to 33% during the morning and evening hours when generation is low And decreases (below 5%) when the generation is high.

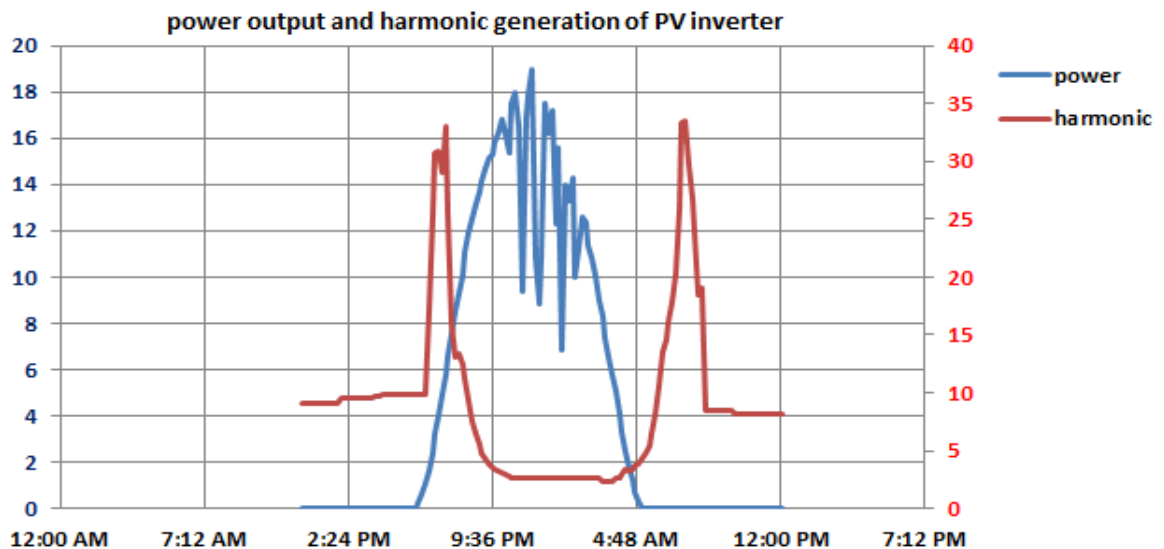


Fig.4.19. Correlation between the generated power and the total current harmonic distortion factor THD_i for 11/03/2014

Also when the PV generation is affected by sudden changes (like passing of clouds) during the day, current THDi is also high. Current harmonic emission of a PV inverter is dependent on PV output power. When generation is higher, the THDi is reduced below the value of 5%. The shape of the THD% compared with the power output in figure (4.19). Very high value of harmonic distortion of the PV current in the early mornings and sunset is due to the inverter's switching device. This will be discussed.

CHAPTER FIVE

SIMULATION RESULTS

5.1. General

Based on the work achieved, the last step seems to be more justified. This chapter presents how the components of the proposed system are implemented in Matlab/Simulink environments. The dynamic behavior of the system will be investigated, using the Matlab/Simulink tools. The main aim of this thesis is to investigate the impact of current harmonics generated by distributed photovoltaic systems connected to a medium voltage network. Also to analyze the impact and effectiveness of the proposed dead-time compensation method, improvement of distorted phase currents is observed.

5.2 Simulation Results of a Single PV Array

The main aim of this chapter is to investigate the impact of current harmonics generated by distributed photovoltaic systems connected to a medium voltage network. The Fast Fourier Transform (FFT) algorithm is used in this chapter to extract the harmonic content from the PV current waveforms. This facility is readily available within the Powergui environment block for SimPower Systems models. The first part of the simulations concentrates on the analysis of a single PV system feeding into a load point via a connecting transformer. The main motivation behind this experiment is to gain insight into the different factors that affect the harmonic generation of a single PV array. The configuration setup of the PV generator is directly taken from the PV generator connected to the power grid presented in figure (5.1).

Different experimental scenarios have been carried out to characterize the voltage and current distortions under different operation conditions. Matlab/Simulink simulation and field experimental results have been provided. The results showed that the harmonic current emission can be affected by the output power level of PV inverter. In addition, the current and voltage waveforms distortion at the PCC to the grid is significantly increased when the net power flow through the bus is close to zero.

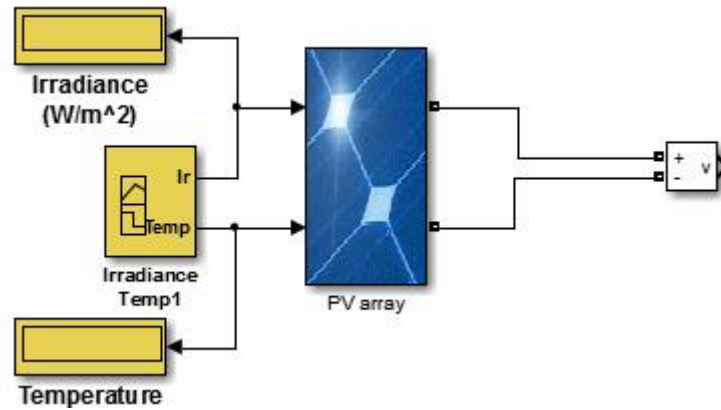


Fig. 5.1. Simulink block of the photovoltaic array

Figure (5.1) shows the mask interface of this PV model. The input parameter for this model is the photovoltaic current. The variation of the current from the photovoltaic varies the photovoltaic output voltage. The Simulink model is derived from the model described in [7].

5.3 Harmonic Currents and Generated Power Simulink

The amount of solar irradiance that hits a photovoltaic array affects the amount of power that the photovoltaic array can produce and supply to the electric grid system. In a typical PV inverter Pulse Width Modulation (PWM) is used to control inverter switches which in turn generate AC output. Although this method of AC generation allows for accurate control of both magnitude and frequency which is important for grid synchronization high order harmonics. In order to reduce the unwanted harmonics compensation techniques have been used for the component of non-linearities of H-bridge inverters used in grid-connected photovoltaic (PV) inverters. Harmonics produced by the electronic control section of the inverter as seen in the inverter layout [18]. Standard IEEE 1547-2003 is the main document which regulates harmonic content from the output of PV systems. The requirement of PV inverters under this standard is that the total harmonic current distortion should be less than 5% of the fundamental component.

The simulation is run at $t=0\text{sec}$ to 0.12sec . at the beginning, the irradiation is set at $G=1000\text{ [W/m}^2\text{]}$ and at $t=0.02\text{sec}$ a step change of irradiation to $600\text{ [W/m}^2\text{]}$

is performed. The output power of the PV varies from 1000W to 7000W of single phase inverter shown in figure(5.2). The PV array operates at maximum power when there is a variation of the irradiance. It is important to mention that the PV simulation modeling in this work does not consider the PV system losses component such as dust factor, module mismatch, cable losses and etc. This gives rise to the difference between the simulated output and the measured output.

The total harmonic distortions of the system without compensation are shown. The current THDi of the system without compensation is 35.0% simulated as shown in Figure (5.2). Higher harmonics are usually caused by non-linear devices in electric power systems, which are widely used in modern power systems, In order to reduce the THDi below 5%, further compensation technique are used to bring it to the required standards. Therefore, the harmonic issue is an important aspect affecting the integration of PV generation.

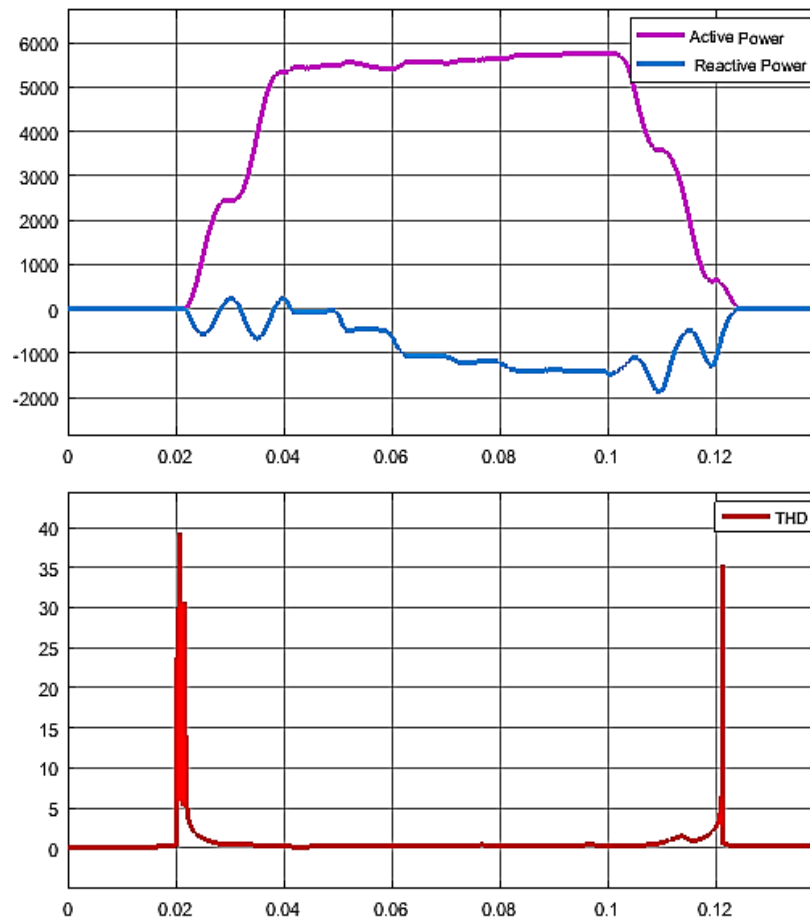


Fig.5.2. power output of inverter and harmonic distortion without compensation technique

5.4 PV Inverter Voltage and Current Output

The simulation results are given in the figure. (5.3) and (5.4) They show the inverter output waveforms: voltage and current (before compensation). Figure(5.3) indicates that the inverter output voltage has a high frequency chopped rectangular waveform (bipolar PWM). In figure. (5.4) the inverter output current, however it follows its reference sine wave. It is clear that, when the inverter output voltage and current passed through a filter, pure sine waves with (220 V, 50 Hz, and 20 A, 50 Hz)These waves can be directly connected to the grid or supply any AC domestic loads.

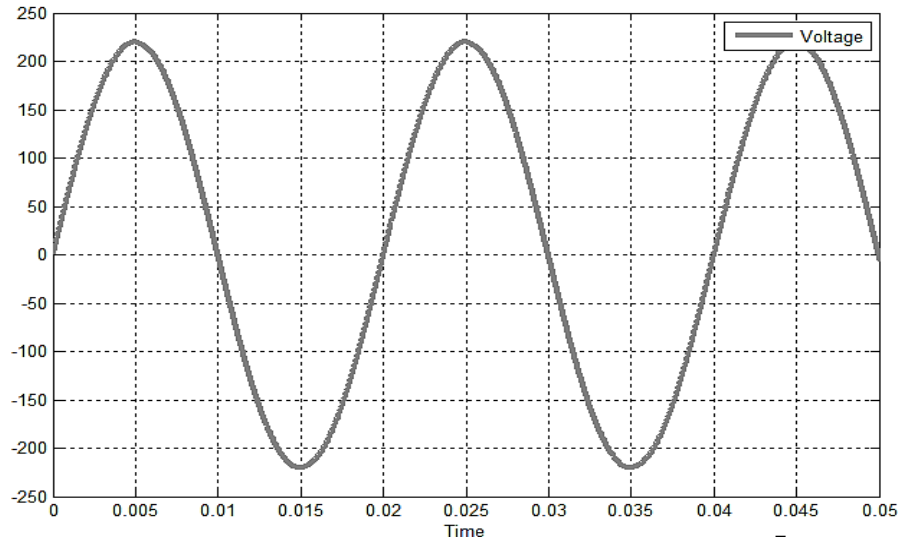


Fig.5.3.voltage output at point of pcc

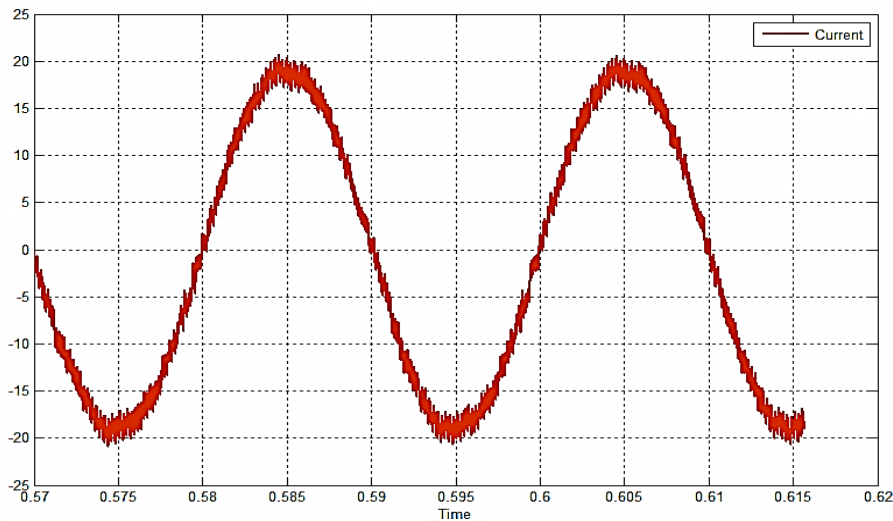


Fig.5.4.
output

current
inverter
Power

5.5

Factor Output Inverter

Most grid connected PV inverters are only set up to inject power at unity power factor, meaning they only produce active power. In fact this decrease the power factor, as the grid is then supplying less active power, but the same amount of reactive power. This problem of poor power factor however can be addressed through the selection of appropriate inverter products. Inverters with reactive power control can be configured to produce both active and reactive power. This means that the power factor for the load can be kept within reasonable limits [13] show in figure(5.5) the system works at unity power factor during the morning and evening when there is low solar output, and solar output is leading PF at peak times, which helps support the voltage.

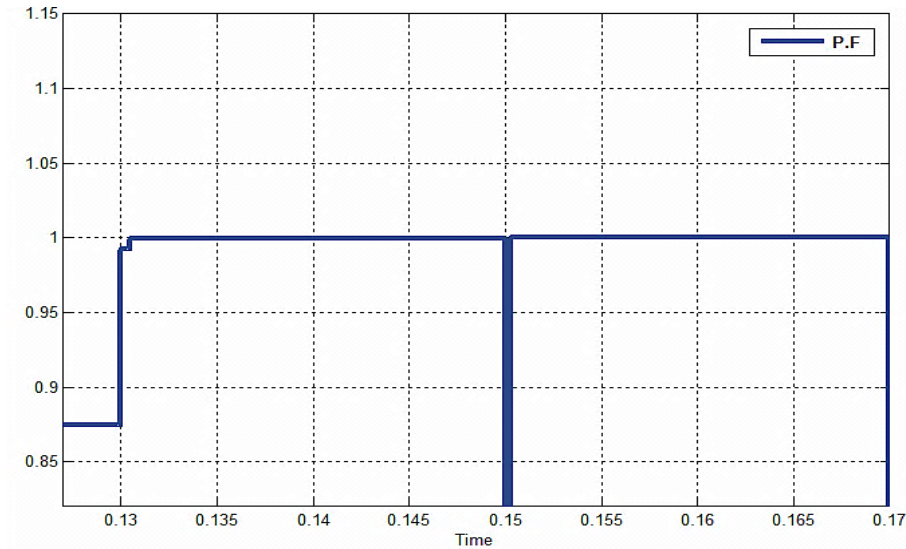


Fig.5.5. power factor

5.6 Discussion

An experimental investigation of the harmonic distortion occurring at the point of common coupling of a number of PV inverters connected to the grid has been carried out on a PV plant of total rated power of 8 kW, single phase. The measured data have been gathered from the installation through a network analyzer with high accuracy for a period of one week.

The analysis of the results show that the current harmonics generated by the PV system highly depend on the generation level. As such, conditions during daylight hours of the generated energy is injected to the grid. On the other hand, during early morning and the sunset the obtained values point out that a small percentage of this energy is injected to the grid. However, this larger harmonic distortion is not harmful when the associated RMS value of the current is relatively low.

There are no DG equipment standards for harmonics in PV inverter[16]. As shown in this section, the harmonic current of PV inverter operate at non-rated condition can be very large. Nowadays, this is not a serious problem to the power quality of the grid since the penetration of PV generation is relatively small. With the increase of PV installation, significant harmonic distortion could come from PV system operating even at low power level. It has been

shown in above section that harmonic current significantly increases when the net current flows through the bus is close to zero. This condition is typical in micro grid with DG and load are near to each other. With consideration of line impedance the voltage quality limits such as EN 50160 may be exceeded.

5.7 The Causes of large THD_i Can be Reduced to The Following:

1. switch frequency (high frequency) current ripple or dead time effect.
2. Grid background voltage distortion
3. DC bus voltage variation.
4. Design of control loop and filter.
5. Measurement resolution or quantitation effects.

5.8 Dead Time Effect on Current Distortions

Dead time distortion (DTD) is a type of cross-over distortion caused by dead-time state of PWM power stages. The problem is present in all PWM output drives. The longer dead time used, the more significant distortion becomes.

Dead-time is a blanking time period (upper & lower transistors in off-state simultaneously) of half-bridge power stage. Both transistors are switched off for short period of time to prevent both transistors conducting simultaneously thus causing a short circuit from supply to ground. To prevent that, a deliberate dead time is always introduced[14].

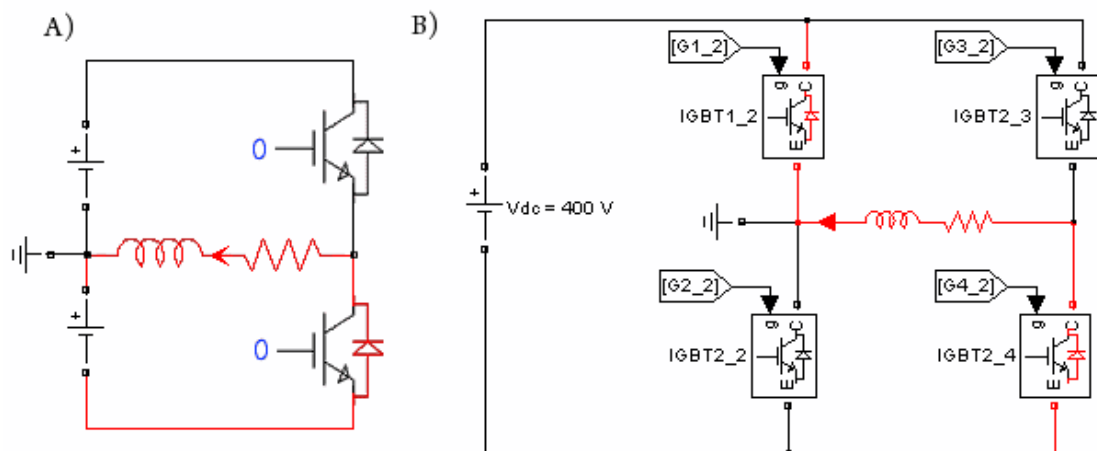


Fig.5.6. Dead time of a half-bridge and a full H-bridge inverter

The dead time compensation is done by a simple edge delay circuit and the phase lag introduced by the compensation method is removed by a simple modification to the reference wave. Its length depends on the switching characteristics of your transistors and your driving circuitry. During dead time, among the active power devices, only the freewheeling diodes conduct. During dead-time, the half- and full-bridge is in a state shown in figure (5.7), [10].

Dead time does introduce a deviation from the correct PWM waveform, and dead time introduces a constant change in that length. The effects of dead time are unwelcome, but it is pretty negligible compared to the alternative. This is why, when designing an inverter, one of the more important tasks is to minimize dead-time while not stepping on some of the limitations of your components and circuitry. In order to avoid shoot-through in the DC/AC inverter, a blanking time is introduced between the turn-off and the turn-on signals. The blanking time was found to be $0.4\ \mu\text{s}$. The DC-AC inverter must be interfaced to the control circuit. includes power supplies and command signals for the inverters.

The control circuit This includes IC to combine shutdown and blanking-time input (the blanking time is adjustable from $0.4\ \mu\text{s}$ to $3.1\ \mu\text{s}$), to drive IC must include a shutdown input. The control input and its current capabilities must be high enough to ensure fast switching. which makes it an expensive solution[15]. The bootstrap drive is therefore selected, since it is cheap, flexible and offers full gate control for sufficient time. The blanking time changes the output voltage from the inverter, and introduces low order current harmonics distortion in the grid, as seen in figure(5.7).

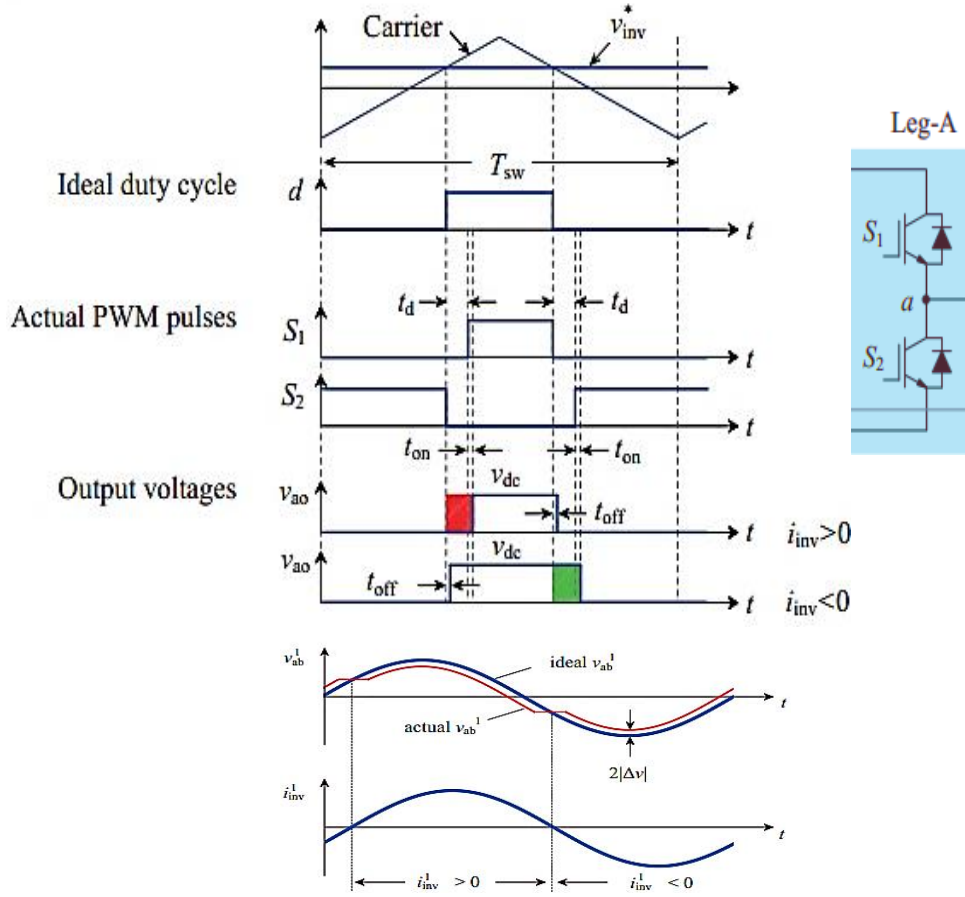


Fig.5.7. Output voltages of Leg-A and considering the dead time effect in one switching period.

5.9 Comparison of Results:

5.9.1 Harmonics and Power Generation Result

The harmonics injected to the particular grid is measured using Fast Fourier Transform (FFT) tool of Simscape Electrical (formerly SimPower Systems and SimElectronics) provides component libraries for modeling and simulating electronic, mechatronic, and electrical power systems. In order to show the harmonic content due to integration of PV system, the built in PV dynamic models has been integrated in to each of the load bus of the grid. To shows the generated power and reactive power and THD, during daily maximum solar irradiance. In addition, the percentage of current THD (THD%) as in figure

(5.8) shows the value of 5% during early of the day and increases to 9% in the afternoon.

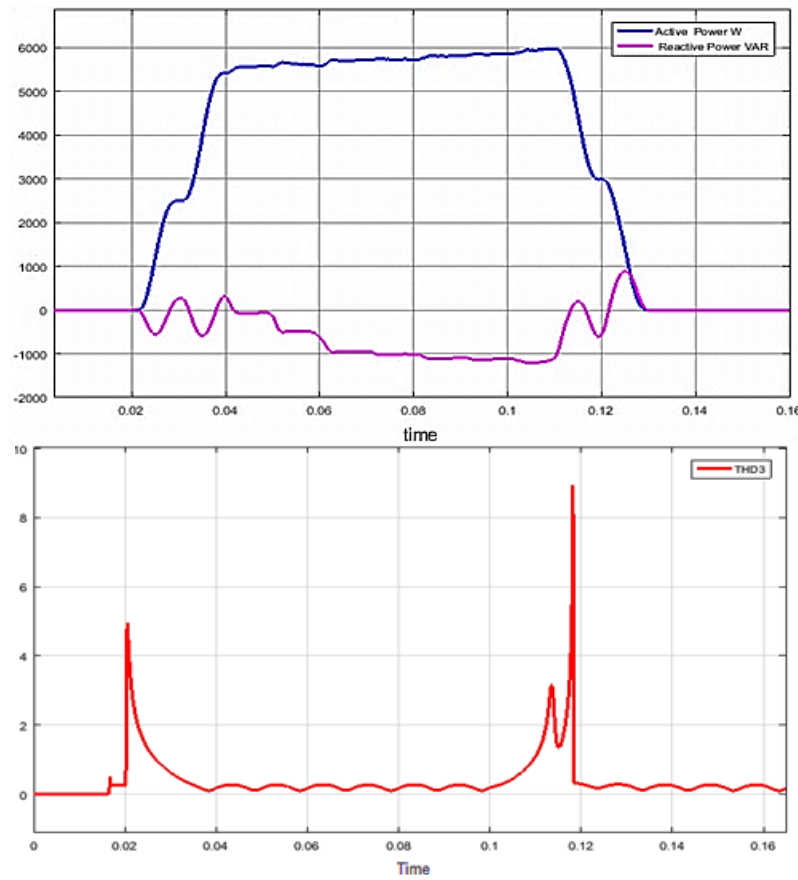


Fig.5.8. power output of inverter and THDi with the compensation technique

Reports of total active power and apparent power measured in time domain. While active power has a behavior like what was expected for this kind of system, it is evident that a remarkable unforeseen oscillation in the harmonic that is preminent especially at the beginning and at the end of the production. It should be remarked that such a high switching frequency for the DC/AC power inverter is not practical because of high switching losses accrued. FFT analysis carried out to find the THDi of the proposed system.

5.9.2 Total Harmonic Distortion (THD) and Current Distortion:

In order to verify the validity of the proposed method, simulations were carried out in MATLAB Simulink environment. The dead time compensation method

proposed in this thesis provides a simple solution for the dead time distortion problem for PWM based single phase inverters. The dead time compensation is done by modification in PWM generation was derived edge delay circuit and the phase lag introduced by the compensation method is removed by a simple modification wave. To analyze the impact and effectiveness of the proposed dead-time compensation method, improved of distorted phase currents is observed. conclude that dead time introduces waveform distortion as well as harmonics at the output current is shown in Figure (5.9). Compensated steady state current waveform of phase A using the proposed method is shown in Figure (5.11). In order to investigate the harmonic spectrum the single sided frequency spectrum of phase A current is also taken in to consideration Figure (5.10),(5.12).

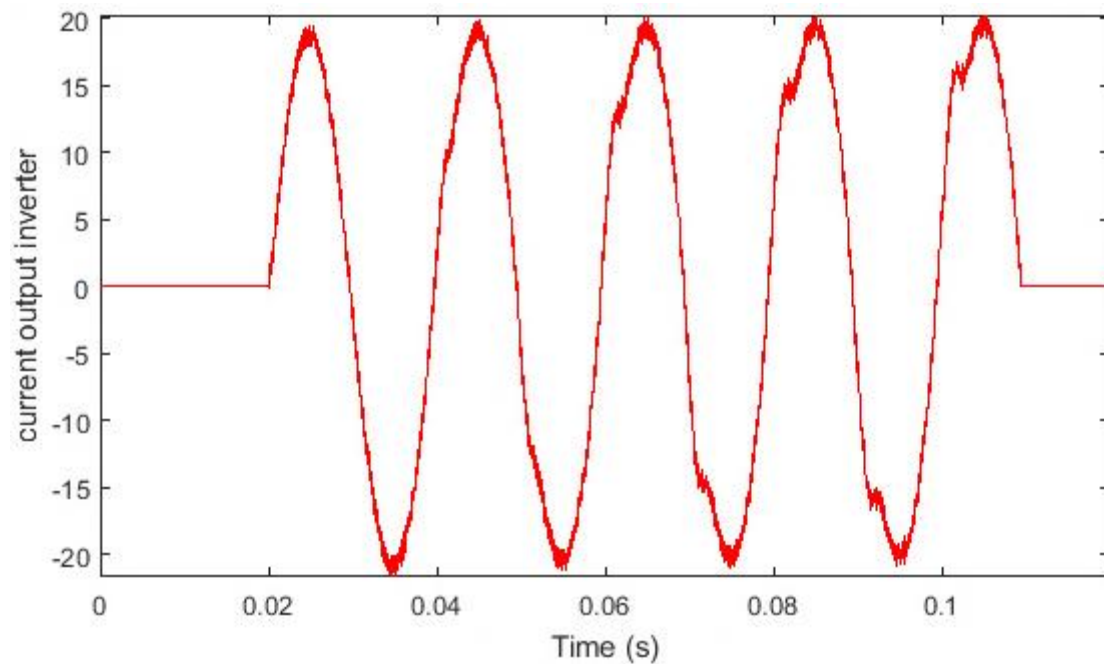


Fig.5.9: Distorted current waveform before compensation

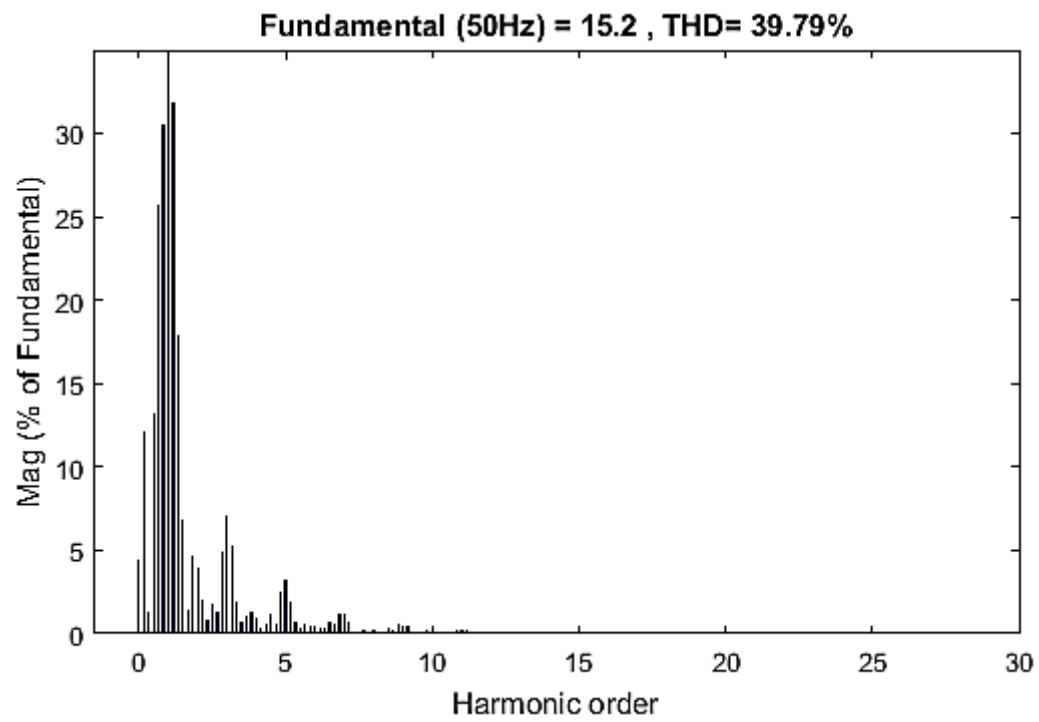


Fig.5.10:THD value of distorted before compensation

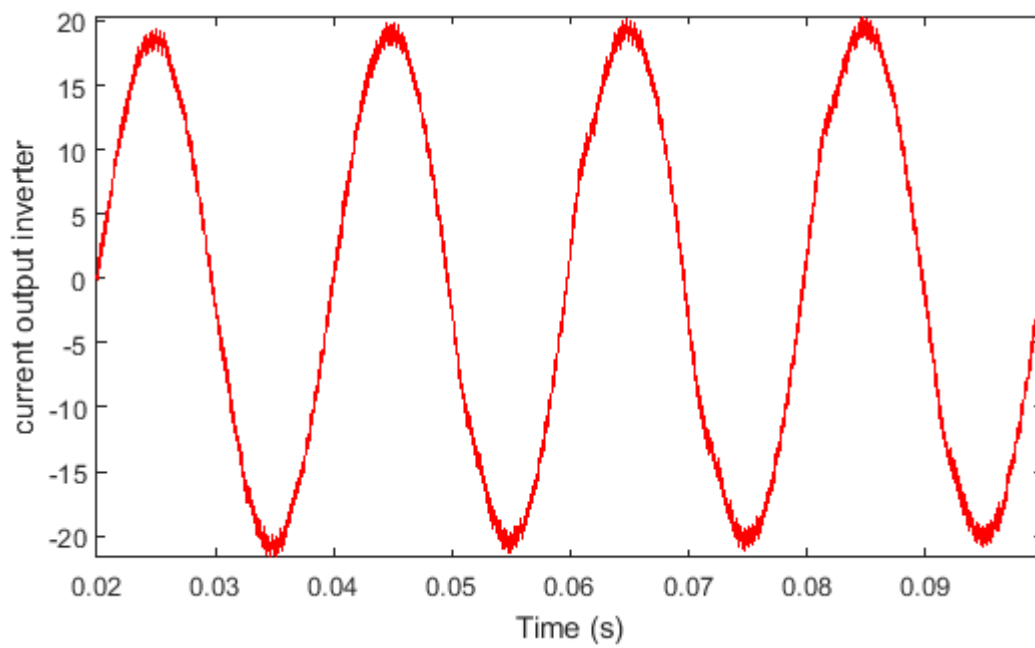


Fig.5.11: Compensated current waveform after compensation

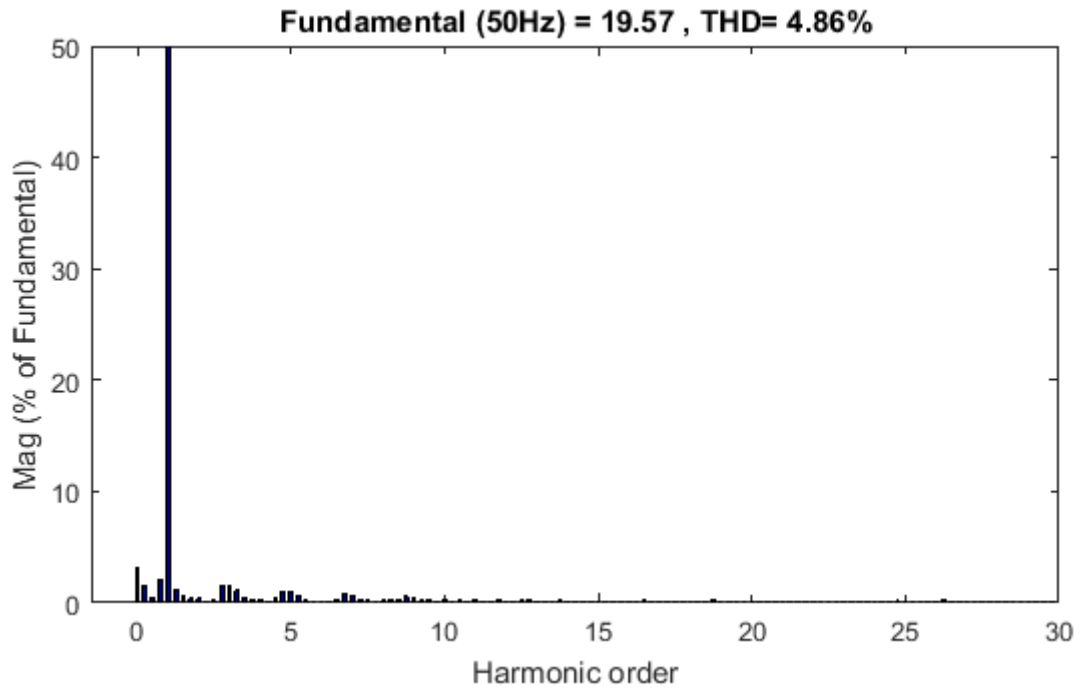


Fig.5.12:THD value of compensated after compensation

By performing Fast Fourier Transform(FFT) at the rated power condition(i.e. this mean applying the standard irradiance 1000 W/m^2 , at the input of the simulated PV system), for 5 cycle of the simulated inverter current, the result of it before and after compensation are shown in Figure (5.10),(5.12), where the $\text{THD}_i\%$ for the inverter current before compensation is 39.79%, after compensation it is a 4.86% which is compatible with the standard[17].The compensation techniques introduced was evaluated by simulations which proved to be effective solution and very satisfactory based on the results obtained.

The compensation method accurately compensates the dead time distortion and the output waveforms are free of any phase delay errors. The validity of the method is confirmed by the simulation results. Current waveforms of compensated output and uncompensated output are provided in order to illustrate the improvement by the compensation method. Frequency spectrums of the distorted and compensated phase current waveforms are provided in order to investigate the harmonic mitigation.

A grid connected single phase PV inverter has been proposed, for ensuring high quality of the current injected to the grid. The compensation technique controller was implemented to reduce the dead time of inverter and lower order harmonics. By using MATLAB simulation it was observed that the THD% value of the PV system is reduced less than 5%.

In conclusion, the proposed dead-time compensation method compensates the magnitude of THDi as well as the phase accurately. Moreover the PWM time compensation provides less harmonic distortion.

CHAPTER SIX

CONCLUSION AND FURTHER WORK

6.1. Conclusion

This thesis dealt with the reduction of current harmonics generated by grid-connected inverters to make the inverter complaint with standard regulations. The solar panels start to generate solar energy as soon as the sunlight hits them in the morning and stops producing solar energy when the sun goes down in the late afternoon. To represent irradiance and also the turn on and turn off times of inverter switching and its effect on the power quality measurements. It has been found that a low solar irradiance (early morning and sunset) has a significant impact of the power quality of the output on the PV system using the THD_i values as a first indication on the impact of the PV system operation on the harmonics. The behavior of the THD_i during the assessment time (i.e. daylight hours), clearly shows that cases with high values of THD_i are limited to the periods with very small solar irradiance (at the sunset and at the sunrise), while the THD is relatively small when there is relatively high solar irradiance. The result shows that voltage THD_i of measured voltage signal is independent of the generated power or the fluctuations of solar irradiance. [8] . On the other hand, the current THD_i value is always exceeding the maximum permissible limit of 5%. To overcome inverter nonlinearity effect proposed method introduced which reduce harmonics by large amount. It can be adapted to varying operating conditions. The dead time compensation method proposed in this thesis provides a simple solution for the dead time distortion problem for PWM .The correction is done by analyzing pulse by pulse in order to correct each pulse individually. The dead time compensation is done by a simple edge delay circuit and the phase lag introduced by the compensation method is removed by a simple modification to the reference wave. PWM switching pulses are obtained by voltage injection.

The possible effects of these harmonic deformation of early morning and sunset are :

- turn on and turn off times of switching device and the increase in frequencies of the photovoltaic inverters caused of (dead time).
- control ring design, filter.
- grid background voltage distortion.

The compensation method accurately compensates the dead time distortion and the output waveforms are free of any phase delay errors. The validity of the method is confirmed by the simulation results. Current waveforms of compensated output and uncompensated output are provided in order to illustrate the improvement by the compensation method. Frequency spectrums of the distorted and compensated phase current waveforms are provided in order to investigate the harmonic mitigation.

In conclusion, the proposed dead-time compensation method compensates magnitude of THD_i as well as the phase accurately. Moreover the PWM provides less harmonic distortion, less switching losses as well as a wide linear range and the output waveforms are free of any phase delay errors.

Apart of this thesis activity was presented in the International Conference on Technical Sciences (ICTS) 2019 which held in Tripoli.

6.2. Future Work:

Any scientific research can't cover all aspects and investigates of all important points at once; some points need to be reinvestigated and studied in future works. Separately from the studies applied in the course of this thesis work; that seems to be of the highest importance, further studies of the impact of the solar systems on the connected power systems and distribution stations needs to be penetrated. Future works that are recommended in other stages of this work are:

- Further investigations of different controllers for the grid current, e.g. the inverter controller and repetitive control for cancellation of harmonic currents are interesting. e.g The effect of compensating the blanking time

which is applied in this research, to get a better grid performance in terms of lower harmonics, etc.

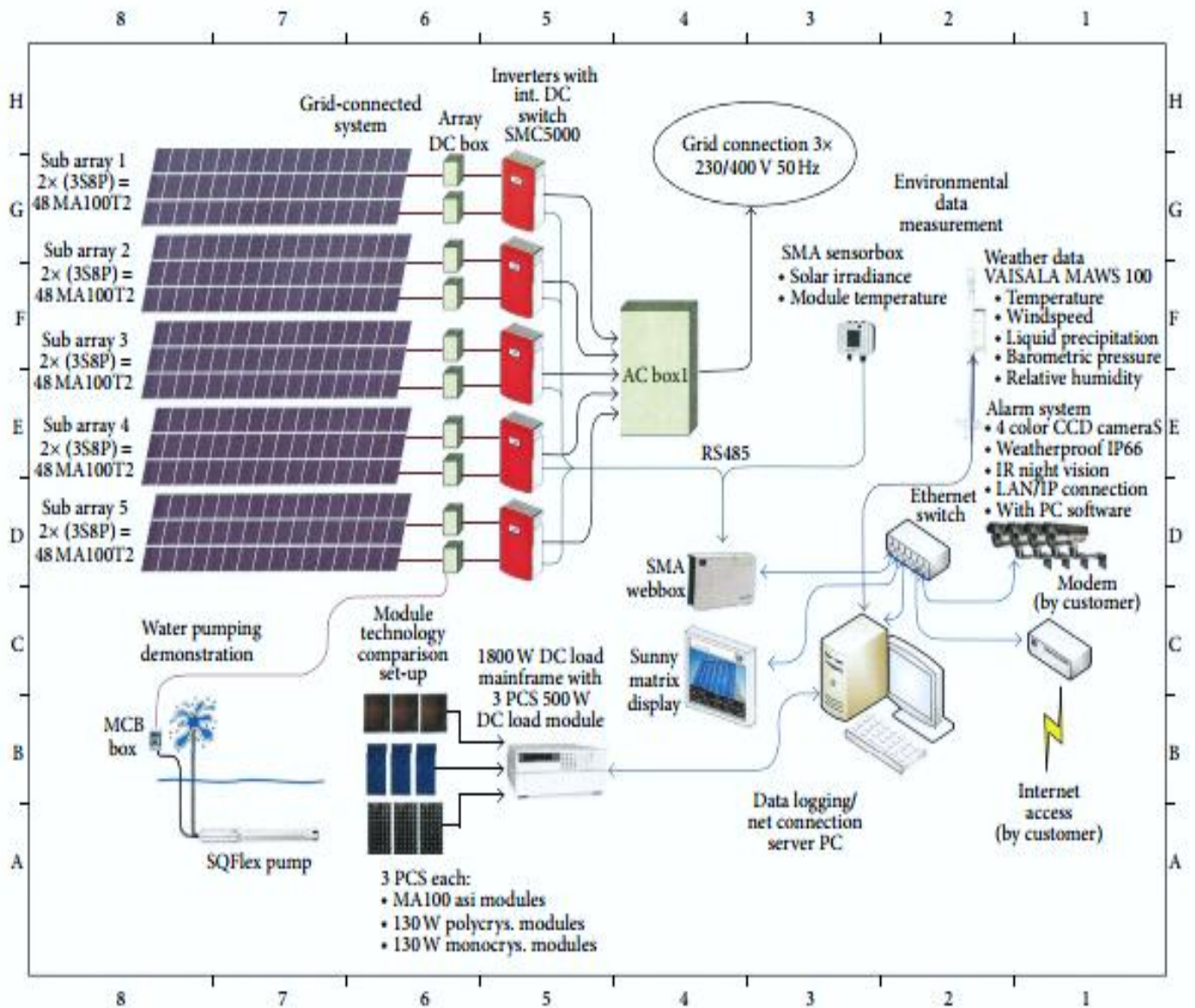
- A more in-depth investigation of grid-connected PV output response for a given variation in solar irradiance.
- Engineering solutions for the handful of adverse effects that a large drop in PV generated power has on the electric grid.
- The reliability analysis on the distribution station of the security and adequacy of the proposed system especially keeping in mind the intermittent function of PV stations.
- Further studies on the protection profile after the injection of PV generators.
- The frequency stability of the power station under the effects of PV systems.

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Appendix A



Appendix B

Technical Data AC/DC Inverter

Specifications of inverters:

Sunny Mini Central 5000A

AC Output

Rated power at 230 V, 50 Hz	5,000 W
Maximum apparent AC power	5,500 VA
Rated grid voltage	230 V
AC nominal voltage	220 V / 230 V / 240 V
AC voltage range	160 V ... 265 V
Nominal AC current at 220 V	22.7 A
Nominal AC current at 230 V	21.7 A
Nominal AC current at 240 V	20.8 A
Maximum output current	26 A
Total harmonic factor of the output current with AC voltage < 2% and AC power > 50% of the rated power	≤ 3%
Rated mains frequency	50 Hz
AC mains frequency	50 Hz/60 Hz
Operating range at AC mains frequency of 50 Hz	44 Hz ... 55 Hz
Operating range at AC mains frequency of 60 Hz	54 Hz ... 65 Hz
Displacement power factor, adjustable*	0.8 _{overexcited} ... 0.8 _{underexcited}
Power factor at nominal AC power**	1
Feed-in phases	1
Connection phases	1
Overvoltage category according to IEC 60664-1	III

* Applies to SMC 5000A-11

** Applies to SMC 5000A

DC input

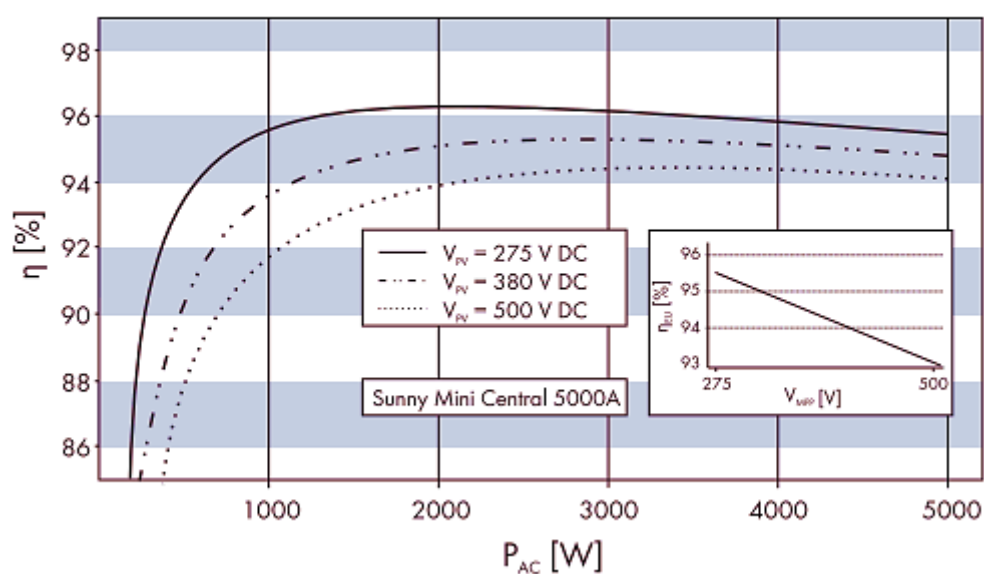
Maximum DC power at $\cos \varphi = 1$	5,750 W
Maximum input voltage*	600 V
MPP voltage range	246 V ... 480 V
Rated input voltage	246 V
Minimum input voltage	211 V
Start input voltage	300 V
Maximum input current	26 A
Maximum input current per string	26 A
Number of independent MPP inputs	1
Strings per MPP input	4

* The maximum open-circuit voltage that can occur at a cell temperature of -10°C must not exceed the maximum input voltage.

Efficiency

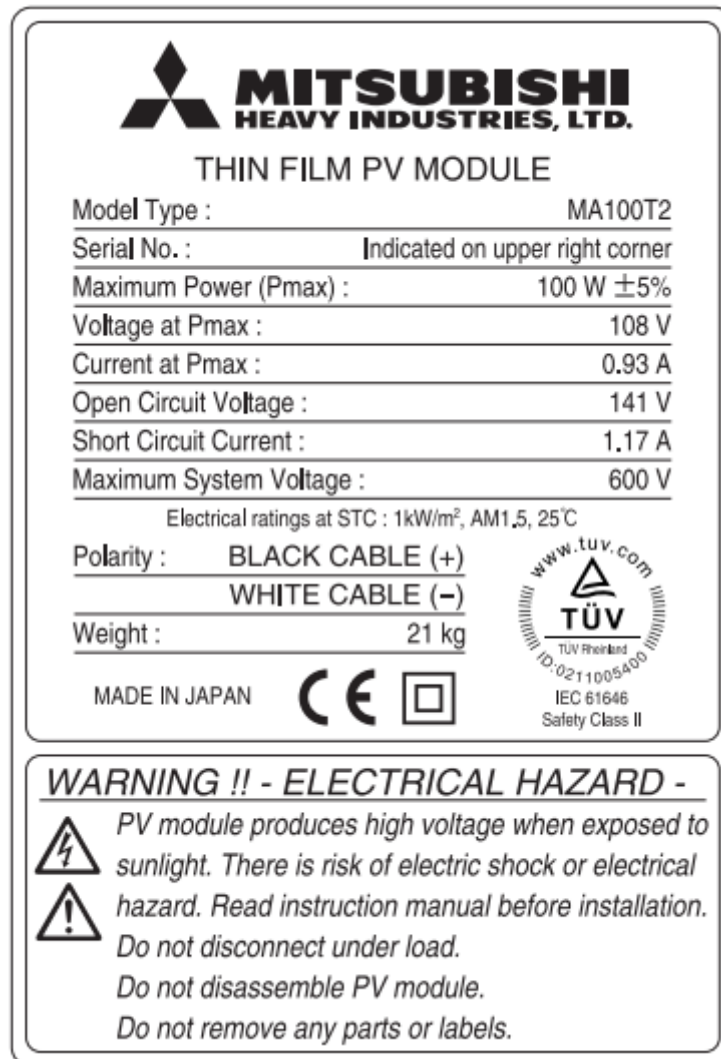
Maximum efficiency, $\eta_{\max}$	96.1%
European weighted efficiency, η_{EU}	95.3%

Efficiency



Appendix C Labeling

- Name Label (75mm× 110mm).



- Serial Number Label (45mm× 20mm)



